Qur’anic Truth-Seeking & Interpretation-Evaluation Framework

A Comprehensive Research and System Design Document

Version 2.0 — Two-Pass Architecture

Classification: Internal Research Document

Audience: Analytically sophisticated practitioners — Bayesian epistemology, knowledge graph architecture, Islamic scholarship

Status: Implementation-Ready (Governance Approved)

Revision History: v1.0 (Apr 5, 2026) Foundational Architecture | v2.0 (Apr 6, 2026) Two-Pass Scoring, Chronological Axis, Structural Skeptic

This document is a system design specification, not a theological treatise. It treats the Qur’an as the subject domain of a structured knowledge system.

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Version: 2.1 — Two-Pass Architecture + Retrieve-Don’t-Generate (RDG)

Classification: Internal Research Document

Audience: Analytically sophisticated practitioners — Bayesian epistemology background, knowledge graph architecture, Islamic scholarship

Status: Implementation-Ready (Governance Approved)

Revision History: v1.0 (Apr 5, 2026) Foundational Architecture | v2.0 (Apr 6, 2026) Two-Pass Scoring, Chronological Axis, Structural Skeptic | v2.1 (Apr 6, 2026) Retrieve-Don’t-Generate Principle, Citation Verification Gate, Hallucination Detection Benchmark

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> Framing Note. This document is a system design specification, not a theological treatise. It treats the Qur'an as the subject domain of a structured knowledge system. It does not make theological claims. All assertions about divine authority, authenticity, or salvific status belong outside this document's scope; what this document does is design a rigorous, auditable framework for evaluating the evidential quality of interpretive claims about the Qur'anic text.

Section 1: Direct Answer — What This Framework Is and Is Not

1.1 The Operational Definition of "Truth" in This Project

The word "truth" in ordinary theological discourse carries salvific, ontological, and metaphysical weight that this framework deliberately does not adjudicate. For operational purposes, this framework adopts the following definition:

> Interpretive truth is the property of an interpretive claim about a Qur'anic passage being well-supported by the most reliable available evidence, coherent with the textual structure of the Qur'an, consistent with the documented understanding of the Arabic in which it was revealed, and traceable to a chain of scholarly reasoning that can be audited, challenged, and revised.

This is a procedural, epistemic definition. It does not assert that a "well-supported" interpretation is what God intended. It asserts that among all interpretations that have been advanced for a given passage, some are better evidenced than others, and that this gradation can be systematized and

exposed transparently.

1.2 Three Distinct Types of Truth — and Why They Must Not Be Conflated

Type	Definition	What This Framework Handles	What It Does Not Handle
Textual Truth	What the Arabic words literally mean, given all known morphological, syntactic, and lexical knowledge	Yes — fully within scope; automatable at high accuracy	No theological claim attached
Interpretive Truth	What scholars across history have argued those words mean in context — the evidence-ranked discourse of tafsir	Yes — core purpose of the system	The system ranks evidence; it does not adjudicate between equally-evidenced competing positions
Theological Truth	What God intended; which interpretation is correct in an ultimate or soteriological sense	Explicitly out of scope	This is a matter of personal belief, scholarly ijtihad, and religious authority — not of data systems

The critical discipline the system must enforce: prevent any output from sliding from Type 1 or 2 claims into Type 3 claims. A score of 92/100 means well-evidenced by classical standards, not divinely authorized.

1.3 What the System Can Do

- Parse and annotate every word of the Qur'anic text with full morphological, lexical, and syntactic metadata
- Retrieve the full evidence trail for a given interpretation across all seven evidence tiers
- Score a user-submitted interpretation across five measurable dimensions and produce a composite Alignment Score
- Flag specific misalignments — lexical, contextual, transmissional, scholarly, or internal — in plain language
- Surface alternative, better-evidenced interpretations from the knowledge graph
- Distinguish between interpretations on which scholars have been historically unanimous, those on which there is recorded disagreement, and those that are modern novelties
- Explain its reasoning at every scoring step with source citations

- Generate two independent scores — a Pass 1 (Textual Logic) score based solely on the Qur'an's internal evidence, and a Pass 2 (Integrated Tradition) score incorporating transmitted and scholarly evidence
- Flag structural divergence between text-internal and tradition-based readings when the delta exceeds 25 points
- Issue TREC flags (Text Requests External Clarification) when the Qur'an's own text internally directs to the Prophetic explanation (e.g., 16:44, 59:7, 4:65)

1.4 What the System Cannot Do

- Issue fatwas or legal rulings
- Determine which of two equally-transmitted competing interpretations is "correct"
- Resolve theological disputes (e.g., Ash'ari vs. Mu'tazili readings of divine attributes)
- Replace the judgment of a qualified mujtahid or mufassir
- Guarantee that its NLP pipeline correctly handles rare morphological constructions or archaic usages
- Certify any interpretation as the "true" meaning intended by God
- Serve as an authoritative religious reference for personal practice

1.5 This Is an Auditable Evidence-Ranking System

The closest analogous system in another domain is a legal precedent database combined with a systematic review platform in evidence-based medicine. Like those systems:

- Every claim is linked to a source
- Sources are graded by quality tier
- Users can inspect the evidence chain
- Disagreements are documented, not suppressed
- The system acknowledges its own uncertainty explicitly

The framework is not designed to produce consensus. It is designed to make the existing scholarly discourse, in all its breadth and disagreement, navigable, measurable, and transparent.

v2.0 introduces a two-pass architecture that separates what the text establishes on its own (Pass 1) from what transmitted tradition adds (Pass 2). This allows users to see precisely where the Qur'anic text and the scholarly tradition converge, diverge, or depend on each other — the highest form of interpretive transparency.

Section 2: Truth Framework — The Four-Layer Model

The four layers are not a ranking of whose interpretation is correct. They are a stratification of epistemic types — what kind of claim is being made and what kind of evidence is marshaled for it. Each layer is legitimate. Each has different evidentiary standards and different confidence ceilings.

2.1 Layer 1: Textual Certainty

What it is: The layer of claims about what the Arabic words demonstrably say — morphology, grammar, and documented lexical range. This is the closest the system comes to objective data.

What counts as evidence:

- Root identification (e.g., - - for khalīfa)
- Part-of-speech tagging (ism fā'il, verb, harf, etc.)
- Morphological parsing (case, gender, number, definiteness)
- Syntactic role in the sentence
- Documented semantic range from classical Arabic lexica: Lane's Lexicon (Edward William Lane, 1863–1893), Lisān al-ʿArab (Ibn Manẓūr, d. 711 AH), Tāj al-ʿArūs (al-Zabīdī, d. 1205 AH), al-Qāmūs al-Muḥīṭ (al-Fīrūzābādī, d. 817 AH)
- Qirāʾāt (recitation variants) where they affect word form

How it is documented: Corpus annotation using the Quranic Arabic Corpus (corpus.quran.com) and CAMeL Tools morphological analyzer; human expert review; cross-reference with classical grammarians (Sībawayhī's al-Kitāb, al-Zamakhsharī's al-Mufaṣṣal)

Key limits:

- Arabic is a polysemous language; even at Layer 1, most significant words carry multiple documented senses
- Root identification is usually unambiguous; sense assignment is not
- Grammar resolves some ambiguities (e.g., case endings); rhetoric introduces others (e.g., metaphor, ellipsis)
- Layer 1 can tell you the possible meanings of a word; it cannot on its own tell you which is operative

Confidence band available: High (for morphological facts); Medium-to-High (for lexical sense assignment); the system should never claim certainty about which sense applies without invoking higher layers

2.2 Layer 2: Transmitted Interpretation (Tafsīr bi'l-Maʾthūr)

What it is: The layer of claims based on what was reliably reported about the meaning of a verse or passage by those closest to the revelation: the Prophet Muhammad, the Companions, and the Successors (tābiʿūn).

What counts as evidence:

- Prophetic explanation of a verse (sunnah tafsīriyya) — authenticated hadith with verified isnād
- Companion reports (āthār) — especially Ibn ʿAbbās (d. 68 AH), Ibn Masʿūd (d. 32 AH), Ubayy ibn Kaʿb (d. ~30 AH), ʿĀʾisha, ʿAlī ibn Abī Ṭālib
- Tābiʿūn reports — Mujāhid ibn Jabr (d. 104 AH), Saʿīd ibn Jubayr (d. 95 AH), Qatāda (d. 117 AH), Ikrima (d. 105 AH), ʿAṭāʾ ibn Abī Rabāḥ (d. 114 AH)
- Asbāb al-nuzūl (occasions of revelation) — from al-Wāḥidī's Asbāb al-Nuzūl, Ibn Ḥajar al-ʿAsqalānī, al-Suyūṭī

How it is documented: Isnād chains evaluated against rijāl al-ḥadīth databases (Kutub al-Sitta, al-Mizzī's Tahdhīb al-Kamāl, Ibn Ḥajar's Taqrīb al-Tahdhīb); classification of narrations as ṣaḥīḥ, ḥasan, ḍaʿīf, mawḍūʿ

Key limits:

- The isnād is not a perfect filter; some ṣaḥīḥ-graded reports have been contested by later scholars
- Many verses have no direct Prophetic tafsīr; the absence of transmitted interpretation is not permission to interpret freely
- Companion reports vary in reliability; Ibn ʿAbbās is considered the highest authority in this layer (he is called Turjumān al-Qurʾān, Interpreter of the Qur'an), but some traditions attributed to him are weak
- The distinction between tafsīr that reached the tābiʿūn through authentic chains versus those representing ijmāʿ (scholarly consensus) of the generation requires care

Confidence band available: High (for ṣaḥīḥ Prophetic tafsīr); Medium-High (for authenticated Companion reports); Medium (for tābiʿūn consensus); Low-Medium (for isolated tābiʿūn reports)

2.3 Layer 3: Inferential Interpretation (Tafsīr bi'l-Raʾy)

What it is: Interpretation derived through disciplined scholarly reasoning — ijtihād — by qualified scholars working within the methodological constraints of Islamic hermeneutics. This is not arbitrary opinion; it is structured argument.

Prerequisites for validity (from the classical tradition):

1. Mastery of Arabic language and rhetoric

2. Knowledge of the full Qur'anic text (to check internal coherence)
3. Knowledge of the sciences of the Qur'an (ʿulūm al-Qurʾān)
4. Knowledge of hadith and sunnah
5. Knowledge of fiqh principles and the positions of earlier scholars
6. Absence of contradiction with settled transmitted interpretation

Key classical sources:

- Al-Zamakhsharī (d. 538 AH), al-Kashshāf — Muʿtazilī theological orientation, outstanding rhetorical analysis
- Fakhr al-Dīn al-Rāzī (d. 606 AH), Mafātīḥ al-Ghayb — Ashʿarī theological orientation, encyclopedic rational method
- Ibn ʿĀshūr (d. 1973), al-Taḥrīr wa'l-Tanwīr — modern classical synthesis
- Al-Alūsī (d. 1854), Rūḥ al-Maʿānī

How it is documented: Citation of the specific reasoning chain; documentation of the assumptions used; record of dissenting views; classification by school (Ḥanafī, Mālikī, Shāfiʿī, Ḥanbalī, Muʿtazilī, Māturīdī, etc.)

Key limits:

- Layer 3 claims can look identical to Layer 2 claims in plain text; the provenance metadata is essential
- The theological school of the interpreter significantly shapes inference; the system must tag this explicitly
- Layer 3 is where most of the historical scholarly disagreement lives; the system should be especially cautious about claiming certainty here
- The line between "disciplined ijtihād" and "speculation dressed as scholarship" is contested; this is why human expert review is irreplaceable

Confidence band available: Medium-High (when the inference is explicit, the reasoning is traceable, and multiple scholars reached the same conclusion independently); Medium (for school-specific readings); Low-Medium (for isolated modern scholarly opinions)

2.4 Layer 4: Speculative, Modern, and Contested Interpretation

What it is: Interpretations that go beyond the transmitted record and beyond classical ijtihād — including modern rationalist re-readings, scientific concordance readings, political readings, esoteric (bāṭinī) readings, and novel theological positions.

Examples:

- "Scientific miracle" readings (e.g., asserting that specific verses predict modern physics)

- Political caliphate readings projected backward onto historical terms
- Progressive-modernist re-readings that deliberately depart from classical transmission
- Esoteric readings from Sufi or Shīʿī traditions that claim inner (bāṭin) meanings not attested in the transmitted record
- Orientalist-academic readings that use comparative Semitic linguistics without reference to the Islamic tradition

How it is documented: Author, year, methodology, stated assumptions, departure point from the transmitted tradition, scholarly reception

Key limits:

- Layer 4 is not automatically wrong; some interpretations here may eventually gain scholarly consensus
- Layer 4 interpretations should be labeled as such regardless of how popular they are
- The system must resist the pressure — from users and from training data — to elevate Layer 4 claims to Layer 2 or 3 status

Confidence band available: Low-to-Uncertain

2.5 Confidence Bands

Band	Description	Typical Layer	Assigned When
High	Strong convergent evidence; minimal scholarly disagreement	Layer 1–2	Morphological facts; ṣaḥīḥ Prophetic tafsīr confirmed across multiple chains
Medium-High	Well-documented in primary sources; limited scholarly dissent	Layer 2–3	Dominant companion position; wide classical consensus
Medium	Multiple attested scholarly opinions; no dominant consensus	Layer 2–3	Recorded ikhtilāf (scholarly disagreement) among competent authorities
Low-Medium	Plausible reasoning but thin transmitted support	Layer 3	School-specific reading not widely adopted; late medieval scholarly opinion
Low	Limited sourcing; relies heavily on inference	Layer 3–4	Isolated modern opinion; esoteric reading

Band	Description	Typical Layer	Assigned When
Uncertain	No traceable scholarly support, or directly contradicts more reliable evidence	Layer 4	Novel claim; political reading; unsourced assertion

2.6 Multiplicity of Meaning as a Feature, Not a Bug

Classical Islamic scholarship does not view legitimate interpretive plurality as a defect. The concept of wujūh (permitted meanings) in ʿulūm al-Qurʾān explicitly recognizes that some words in the Qur'an were intended to carry multiple valid senses simultaneously. Al-Suyūṭī's al-Itqān fī ʿUlūm al-Qurʾān catalogs over 80 types of Qur'anic sciences precisely because the text was understood as inexhaustible in its dimensions.

The system operationalizes this by:

1. Never forcing a single "correct" answer when the evidence supports multiple legitimate readings
2. Documenting competing positions as separate nodes in the knowledge graph with their own evidence trails
3. Scoring interpretations relative to their claimed type — a Layer 4 interpretation is scored against Layer 4 standards, not against the high bar of Layer 2
4. Signaling ambiguity explicitly rather than suppressing it behind a single composite score

A verse that the system rates as "3 well-supported interpretations, Medium confidence across all three, no dominant consensus" is not a failure. It is an accurate reflection of the state of the scholarly record.

Section 3: Evidence Hierarchy — Seven Tiers

3.1 The Hierarchy Table

Tier	Source Type	Key Examples	Evidentiary Strength	Key Limits
1	Qur'an explaining Qur'an (tafsīr al-Qurʾān bi'l-Qurʾān)	General statement in one passage specified by another (e.g., general command to pray in surah 2 specified by prayer details elsewhere); shared root usage across passages; thematic parallels	Highest — part of the same revealed text; no isnād required	Requires careful argument that the cross-reference is genuine and not merely lexical coincidence; can be abused to force meaning onto a passage
2	Authenticated Prophetic explanation (tafsīr nabawī)	Ṣaḥīḥ hadith in which the Prophet explains a verse — e.g., the hadith in Ṣaḥīḥ Muslim explaining al-ẓālimīn in Q6:82 as shirk; the Prophet's explanation of al-ṣirāṭ al-mustaqīm in the Fātiḥa	Very High — direct from the recipient of revelation	Only a small fraction of verses have direct Prophetic tafsīr; the isnād must be assessed; some hadith cited as tafsīr are ḥasan or ḍaʿīf

Tier	Source Type	Key Examples	Evidentiary Strength	Key Limits
3	Major Companion tafsīr	Ibn ʿAbbās (d. 68 AH) — Turjumān al-Qurʾān; Ibn Masʿūd (d. 32 AH); Ubayy ibn Kaʿb (d. ~30 AH); ʿAlī ibn Abī Ṭālib; ʿĀʾisha (for verses relevant to household/personal matters); Zayd ibn Thābit	High — witnessed revelation in its context; many learned directly from the Prophet	Some traditions attributed to Ibn ʿAbbās have weak chains; Companions themselves sometimes disagreed; Isrāʾīliyyāt (Judeo-Christian narratives) entered through some Companion channels
4	Early Successor (tābiʿūn) tafsīr	Mujāhid ibn Jabr (student of Ibn ʿAbbās; d. 104 AH); Saʿīd ibn Jubayr (d. 95 AH); Qatāda (d. 117 AH); Ikrima (d. 105 AH); ʿAṭāʾ ibn Abī Rabāḥ (d. 114 AH); al-Ḥasan al-Baṣrī (d. 110 AH); al-Ḍaḥḥāk ibn Muzāḥim	Medium-High — learned from Companions; high scholarly credibility	Generation further from revelation; school affiliations begin to shape interpretation; regional variation (Mecca, Medina, Kufa, Basra schools)

Tier	Source Type	Key Examples	Evidentiary Strength	Key Limits
5	Classical compiled tafsīr works	Al-Ṭabarī (d. 310 AH), Jāmiʿ al-Bayān; Ibn Kathīr (d. 774 AH), Tafsīr al-Qurʾān al-ʿAẓīm; Al-Qurṭubī (d. 671 AH), al-Jāmiʿ li-Aḥkām al-Qurʾān; Al-Baghawī (d. 516 AH), Maʿālim al-Tanzīl; Al-Zamakhsharī (d. 538 AH), al-Kashshāf; Fakhr al-Dīn al-Rāzī (d. 606 AH), Mafātīḥ al-Ghayb; Ibn ʿĀshūr (d. 1973), al-Taḥrīr wa'l-Tanwīr	Medium — systematic compilations; synthesize earlier tiers; each work carries its school's biases	Each work must be read with knowledge of the author's theological and legal school; al-Zamakhsharī's Muʿtazilī orientation shapes his readings of divine attributes; Ibn Kathīr's Ḥanbalī-Salafī orientation shapes his; compiler's hadith evaluation competence varies
6	Classical Arabic philology and lexicography	Sībawayhī (d. ~180 AH), al-Kitāb; Al-Khalīl ibn Aḥmad, Kitāb al-ʿAyn; Al-Zamakhsharī, al-Mufaṣṣal; Ibn Manẓūr (d. 711 AH), Lisān al-ʿArab; Al-Fīrūzābādī (d. 817 AH), al-Qāmūs al-Muḥīṭ; Edward William Lane (d. 1876), An Arabic-English Lexicon; al-Rāghib al-Iṣfahānī (d. ~502 AH), al-Mufradāt fī Gharīb al-Qurʾān (specifically Qur'anic vocabulary)	Medium — essential for establishing the lexical range of Arabic words; grounds Tier 1 analysis	Lexicons document possible senses, not operative sense in a given verse; pre-Islamic poetry usage may not transfer to Qur'anic usage; Lane relies heavily on classical sources and should be cross-checked

Tier	Source Type	Key Examples	Evidentiary Strength	Key Limits
7	Modern scholarly analysis	Contemporary academic tafsīr; Fazlur Rahman, Major Themes of the Qurʾan; Seyyed Hossein Nasr (ed.), The Study Quran; Angelika Neuwirth's structural analysis; comparative Semitic philology; corpus linguistics applied to Qur'anic Arabic; archaeological and historical contextualization	Low-Medium — can surface new linguistic or historical data; often methodologically innovative	Wide variation in methodology and scholarly accountability; some works operate from secular-critical assumptions incompatible with traditional tafsīr standards; others from specific modern theological agendas; all must be treated as Layer 3–4 sources unless they are explicitly synthesizing higher tiers

3.2 Where Each Tier Applies

Tier 1 is applicable whenever the Qur'an makes a thematic statement on the same topic elsewhere with a shared lexical field. It is most powerful for: (a) passages where a general pronoun or undefined term is specified elsewhere; (b) recurring Qur'anic vocabulary where established patterns of usage constrain meaning; (c) structural/ring-composition arguments about a surah's internal coherence. It must be distinguished from mere verbal parallelism.

Tier 2 is decisive when it exists and is authentic. However, it covers relatively few verses directly, and hadith scholarship requires specialized competence that general users lack. The system must auto-flag any Tier 2 citation with its hadith grade.

Tier 3 is the most commonly invoked tier in classical tafsīr and carries the weight of transmission from the generation of revelation. Ibn ʿAbbās's positions, in particular, are treated as near-authoritative by the majority of the classical tradition.

Tier 4 adds regional diversity and bridges the Companion era to the compilation era. The Basra, Kufa, and Medina schools sometimes diverge, and these divergences are documented features of the tradition, not errors.

Tier 5 is where most readers encounter the tradition. The key discipline is understanding that al-Ṭabarī synthesizes Tiers 1–4 within his framework; Ibn Kathīr does the same with a more critical hadith lens; al-Qurṭubī foregrounds fiqh; al-Zamakhsharī foregrounds rhetoric.

Tier 6 is foundational but auxiliary. No interpretation is valid if it violates established Arabic morphology or syntax. But lexical possibility alone does not make an interpretation valid.

Tier 7 should be cited as supplementary, not primary. Its most legitimate use is: (a) introducing new data (archaeological, papyrological, historical) not available to classical scholars; (b) systematic linguistic analysis that applies formal methods; (c) comparative analysis with pre-Islamic Arabic poetry to establish semantic range.

3.3 Common Misuse Patterns

Misuse	Tier Affected	Description
Tier-jumping	All	Citing a Tier 5 or 7 source as if it overrides Tier 2 consensus
Cherry-picking weak isnāds	Tier 2–3	Selecting a ḍaʿīf (weak) or mawḍūʿ (fabricated) hadith that supports a preferred reading while ignoring ṣaḥīḥ reports that contradict it
Isrāʾīliyyāt laundering	Tier 3–5	Treating narratives of Jewish or Christian origin that entered classical tafsīr through some Companion channels as transmitted Islamic interpretation
Lexical absolutism	Tier 6	Treating Lane's Lexicon or Lisān al-ʿArab as providing a single correct meaning and applying it decontextually
Modern authority inflation	Tier 7	Treating a 20th-century scholar's opinion as equivalent in weight to classical transmitted interpretation
Qirāʾāt neglect	All	Ignoring the fact that a variant recitation changes the morphological form of a word, and thus its lexical or grammatical role

Misuse	Tier Affected	Description
Occasion-of-revelation overgeneralization	All	Treating the specific historical context of revelation (asbāb al-nuzūl) as a limiting factor on a verse's application, when classical scholars held the rule: al-ʿibra bi-ʿumūm al-lafẓ lā bi-khuṣūṣ al-sabab (the lesson is the generality of the wording, not the specificity of the occasion)

Section 4: Persona Roster — Human Expert Roles Required in Reality

The following table specifies every expert role required for a credible real-world build of this framework. "What fails without them" is the failure mode most likely to emerge in an AI-only or under-staffed build.

#	Role Title	Why Essential (What Fails Without Them)	Required Qualifications	Approx. FTE %
1	Qur'anic Methodology Lead	Without a methodology gatekeeper, the system will conflate the four layers and produce pseudo-authoritative scores on genuinely contested matters	PhD in Qur'anic studies or Islamic theology; deep familiarity with uṣūl al-tafsīr; ability to arbitrate between scholars on methodological disputes	15%
2	Classical Tafsīr Scholar (Sunni mainstream)	Without someone who can read and audit classical Arabic tafsīr directly, the system will misclassify, misquote, or fail to detect anachronistic readings of classical sources	Advanced training in Islamic sciences (ʿālim-level or equivalent); direct reading knowledge of al-Ṭabarī, Ibn Kathīr, al-Qurṭubī, al-Rāzī, al-Zamakhsharī in Arabic	20%
3	Arabic Linguist / Philologist	Without expert linguistic review, the NLP pipeline will misparse morphologically complex forms, assign incorrect roots, and produce wrong lexical sense assignments	PhD in Arabic linguistics or classical Arabic philology; competence in Sībawayhī's grammar tradition; familiarity with Lisān al-ʿArab and Lane's Lexicon	15%

#	Role Title	Why Essential (What Fails Without Them)	Required Qualifications	Approx. FTE %
4	ʿUlūm al-Qurʾān Specialist	Without someone dedicated to the Qur'anic sciences, the system will mishandle asbāb al-nuzūl, ignore qirāʾāt variants that change meaning, and miss naskh (abrogation) relationships that affect interpretation	Specialization in ʿulūm al-Qurʾān; familiarity with al-Suyūṭī's al-Itqān, al-Zarkashī's al-Burhān, Ibn ʿAqīla's works; knowledge of the canonical ten qirāʾāt	10%
5	Hadith Critic / Isnād Specialist	Without isnād expertise, the system will assign incorrect reliability grades to Tier 2–3 sources, potentially elevating ḍaʿīf or mawḍūʿ narrations	Specialization in ʿilm al-rijāl and ḥadīth criticism; competence with Kutub al-Sitta, al-Mizzī's Tahdhīb al-Kamāl, and modern hadith databases	10%
6	Islamic Legal / Theological Analyst	Without fiqh and ʿaqīdah expertise, the system will conflate legal rulings (aḥkām) derived from a verse with its thematic meaning, and will miss when a "theological reading" is actually a school-specific legal reading in disguise	Training in at least two of the major legal schools (madhāhib); familiarity with uṣūl al-fiqh; knowledge of ʿaqīdah schools (Ashʿarī, Māturīdī, Atharī/Ḥanbalī)	8%

#	Role Title	Why Essential (What Fails Without Them)	Required Qualifications	Approx. FTE %
7	Knowledge Graph Architect	Without graph architecture expertise, the ontology will have ill-defined entity types, ambiguous edge semantics, and no provenance model — producing a database that is fast to build and impossible to trust	Experience with property graph databases (Neo4j) or RDF triple stores; ontology design competence; knowledge of data modeling for human ities/knowledge representation	10%
8	Arabic NLP / ML Scientist	Without ML expertise in Arabic, the NLP pipeline will perform poorly on Qur'anic Classical Arabic, produce wrong morphological analyses for rare forms, and fail to generalize across qirāʾāt variants	PhD or equivalent in NLP/ML; specific competence in Arabic NLP (CAMeL Tools, AraBERT/CAMeLBERT); experience with low-resource classical language NLP	15%
9	Evaluation Scientist / Benchmark Designer	Without evaluation expertise, the scoring rubric will have unmeasured construct validity, the benchmark will be contaminated, and the confidence estimates will be meaningless	Background in psychometrics, information retrieval evaluation, or NLP benchmark design; familiarity with inter-annotator agreement metrics (Cohen's κ, Krippendorff's α)	8%

#	Role Title	Why Essential (What Fails Without Them)	Required Qualifications	Approx. FTE %
10	Product Designer (UX)	Without UX expertise, the framework will produce outputs that expert users cannot navigate and that general users will misread (e.g., treating a "Low" score as a theological rejection)	Competence in information architecture; experience designing for high-stakes, nuanced information environments; ideally some background in educational software	5%
11	Governance / Red-Team Reviewer	Without a governance role, sectarian biases in the training data will go undetected; AI hallucinations of hadith citations will not be caught; politically sensitive outputs will not be reviewed before release	Background in AI ethics, Islamic studies, and governance; ability to audit the system from multiple sectarian and cultural perspectives; independent from the core development team	5%
12	Community Scholar Liaison	Without this role, the project will not gain trust from the Muslim scholarly community, and will not have access to the tacit knowledge that makes classical scholarship coherent	Respected standing in at least one scholarly community (Sunni, Shīʿī, or Ibāḍī); relationships with senior scholars; understanding of how religious authority works institutionally	5%

#	Role Title	Why Essential (What Fails Without Them)	Required Qualifications	Approx. FTE %
13	Structural Skeptic	Without a dedicated adversarial role, the system will fail to detect when a word's assigned meaning in one verse contradicts the theology of another. This persona prevents "Lexical Absolutism" — the error of locking a word into one sense and carrying it across the corpus without checking for cross-verse incoherence	Background in formal logic or analytical philosophy; deep familiarity with the entire Qur'anic text; competence in identifying contradictions, category errors, and over-generalized readings	8%

Total approximate FTE requirement for a production-grade team: 6–9 FTE across these 13 roles

4.2 Subordinated Auditor Roles (v2.0)

In v2.0, Persona #2 (Classical Tafsir Scholar) and Persona #5 (Hadith Critic / Isnād Specialist) serve a dual function:

1. Pass 2 engine: They power the Integrated Tradition scoring layer (D3 and D4)
2. Pass 1 auditors: They review Pass 1 outputs to detect when the "closed-loop" text analysis is silently importing assumptions from the scholarly tradition it claims to exclude

This subordination structure ensures that the tradition is available without being dominant. The text is analyzed first; the tradition is consulted second; and any divergence is made visible.

Section 5: Knowledge Graph Design — Semantic Architecture

5a. Core Entity Types (Node Types)

5a.1 Surah

```
{
"node_type": "Surah",
"properties": {
"surah_id": "integer (1–114)",
"name_arabic": "string",
"name_transliterated": "string",
"name_meaning": "string",
"revelation_type": "enum: Makk■ | Madan■ | Disputed",
"chronological_order": "integer (classical Nöldeke-derived estimate; note: multiple competing
lists exis…
"chronological_order_source": "string (e.g., 'al-Suy■■■ list', 'Ibn ■ajar list')",
"verse_count": "integer",
"word_count": "integer",
"letter_count": "integer",
"themes_primary": "array of Theme node IDs",
"structural_notes": "string (e.g., ring composition, seven-verse division)",
"source_id": "string",
"confidence_score": "float (0–1)",
"last_reviewed": "datetime",
"reviewer_id": "string"
}
}
```

5a.2 Verse (Āya)

```
{
"node_type": "Verse",
"properties": {
"verse_id": "string (e.g., '2:255')",
"surah_id": "integer",
"aya_number": "integer",
"text_arabic": "string (Uthmanic orthography)",
"text_uthmanic_rasm": "string",
"text_transliteration": "string (ALA-LC or Buckwalter)",
"text_translation_en_primary": "string (with translator attribution)",
"word_count": "integer",
"saj■_group": "string (rhyme group label if applicable)",
"structural_position": "string (e.g., 'center of ring in Sura 2', 'opening of hizb')",
"muhkam_or_mutashabih": "enum: Muhkam | Mutashabih | Both | Unclassified",
```

```
"makki_madani_override": "enum: Makk■ | Madan■ | null (inherits from Surah unless
exceptional)",
"revelation_period": "enum: early_makki | middle_makki | late_makki | early_madani |
late_madani | dispu…
"revelation_period_confidence": "enum: High | Medium | Low (how certain is the period
assignment)",
"revelation_period_source": "string (classical source for the dating: e.g., 'al-Suy■■■,
al-Itq■n', 'Nöld…
"source_id": "string",
"confidence_score": "float (0–1)",
"last_reviewed": "datetime",
"reviewer_id": "string"
}
}
```

5a.3 Word

```
{
"node_type": "Word",
"properties": {
"word_id": "string (e.g., '2:255:1' = surah:verse:position)",
"position_in_verse": "integer",
"surface_form": "string (as written in mushaf)",
"surface_form_vocalized": "string (with tashk■l/■arak■t)",
"lemma": "string",
"root": "string (3 or 4 radical letters)",
"POS_tag": "string (Quranic Arabic Corpus tagset: noun, verb, particle, etc.)",
"morphological_features": {
"case": "string (nominative | accusative | genitive | null)",
"number": "string (singular | dual | plural)",
"gender": "string (masculine | feminine)",
"definiteness": "string (definite | indefinite | construct)",
"verb_form": "string (perfect | imperfect | imperative | masdar | participle | null)",
"person": "string (first | second | third | null)",
"voice": "string (active | passive | null)"
},
"syntactic_role": "string (subject | predicate | object | adjective | etc.)",
"semantic_domain_primary": "string (from root semantic field)",
"polysemy_flag": "boolean",
"rare_usage_flag": "boolean",
"source_id": "string",
"confidence_score": "float (0–1)",
"last_reviewed": "datetime",
"reviewer_id": "string"
}
}
```

5a.4 Root

```
{
"node_type": "Root",
"properties": {
"root_id": "string (e.g., '■■■')",
```

```
"root_form": "string (3 or 4 radicals)",
"root_type": "enum: trilateral | quadrilateral",
"semantic_field_primary": "string",
"semantic_field_secondary": "array of strings",
"lisan_alarab_entry": "string (key definitions from Ibn Man■■r)",
"lane_lexicon_entry": "string (key definitions from Lane)",
"alraghib_entry": "string (from *al-Mufrad■t* if Qur'anic root)",
"preislamic_attestation": "boolean",
"quran_occurrence_count": "integer",
"semantic_range_notes": "string",
"source_id": "string",
"confidence_score": "float (0–1)",
"last_reviewed": "datetime",
"reviewer_id": "string"
}
}
```

5a.5 Tafsir_Claim

```
{
"node_type": "Tafsir_Claim",
"properties": {
"claim_id": "string",
"verse_id": "string",
"claim_text_arabic": "string",
"claim_text_english": "string",
"claim_type": "enum: lexical | contextual | legal | theological | allegorical | scientific |
thematic",
"layer": "integer (1–4, per Section 2)",
"tier": "integer (1–7, per Section 3)",
"source_scholar_id": "string (Scholar node ID)",
"source_work": "string (title of tafs■r or primary work)",
"source_page_or_volume": "string",
"school": "string (e.g., '■anbal■', 'Ash■ar■', 'Mu■tazil■', 'M■lik■', 'Modern Academic')",
"language": "string",
"consensus_status": "enum: ijm■■ (consensus) | mashh■r (well-known) | marj■■ (outweighed) |
shadhdh (ano…
"confidence_score": "float (0–1)",
"date_entered": "datetime",
"last_reviewed": "datetime",
"reviewer_id": "string",
"source_id": "string"
}
}
```

5a.6 Scholar

```
{
"node_type": "Scholar",
"properties": {
"scholar_id": "string",
"name_arabic": "string",
"name_transliterated": "string",
```

```
"birth_year_AH": "integer",
"death_year_AH": "integer",
"era": "enum: Companion | Successor | Early Classical | Classical | Late Classical | Modern",
"legal_school": "string",
"theological_school": "string",
"primary_works": "array of strings",
"tafsir_methodology": "string (brief characterization)",
"reliability_assessment": "string (for hadith transmitters: rij■l evaluation)",
"source_id": "string",
"confidence_score": "float (0–1)"
}
}
```

5a.7 Hadith

```
{
"node_type": "Hadith",
"properties": {
"hadith_id": "string",
"matn_arabic": "string (text of the hadith)",
"matn_english": "string",
"sanad_summary": "string (key transmitters)",
"grade": "enum: ■a■■■ | ■asan | ■a■■f | maw■■■ | l■ a■la_lahu",
"grade_source": "string (who graded it and in which work)",
"kitab": "string (e.g., '■a■■■ al-Bukh■r■')",
"book_number": "integer",
"hadith_number": "integer",
"chapter": "string",
"tafsir_relevance": "array of Verse IDs",
"source_id": "string",
"confidence_score": "float (0–1)",
"last_reviewed": "datetime",
"reviewer_id": "string"
}
}
```

5a.8 Asbab_al_Nuzul

```
{
"node_type": "Asbab_al_Nuzul",
"properties": {
"occasion_id": "string",
"verse_id": "string",
"narration_text": "string",
"narrator_chain": "string",
"narrator_reliability": "string",
"primary_source_work": "string (e.g., 'al-W■■id■, Asb■b al-Nuz■l')",
"reliability_grade": "enum: ■a■■■ | ■asan | ■a■■f | disputed",
"historical_context": "string",
"hermeneutical_implication": "string (how this changes or constrains interpretation)",
"source_id": "string",
"confidence_score": "float (0–1)",
"last_reviewed": "datetime"
```

```
}
}
```

5a.9 Qirāʾa (Recitation Variant)

```
{
"node_type": "Qiraa",
"properties": {
"qiraa_id": "string",
"word_id": "string",
"variant_form": "string (how the word is read in this recitation)",
"reciter_authority": "string (e.g., '■af■ ■an ■■■im', 'Warsh ■an N■fi■')",
"authority_chain_grade": "enum: mutaw■tir | mashh■r | ■■■d",
"morphological_impact": "string (how the variant changes POS or case)",
"semantic_impact": "string (how the variant changes meaning)",
"ruling_impact": "boolean (does this variant affect a legal ruling?)",
"classical_discussion": "string (summary of scholarly debate about this variant)",
"source_id": "string",
"confidence_score": "float (0–1)"
}
}
```

5a.10 Rhetorical_Device

```
{
"node_type": "Rhetorical_Device",
"properties": {
"device_id": "string",
"verse_id": "string",
"device_type": "string (e.g., tashb■h/simile, isti■■ra/metaphor, kin■ya/metonymy,
iltif■t/shift of addre…
"location_words": "array of Word IDs",
"description": "string",
"hermeneutical_relevance": "string (how awareness of this device changes interpretation)",
"classical_discussion_source": "string",
"source_id": "string",
"confidence_score": "float (0–1)"
}
}
```

5a.11 Theme

```
{
"node_type": "Theme",
"properties": {
"theme_id": "string",
"name_arabic": "string",
"name_english": "string",
"cluster": "string (e.g., 'divine attributes', 'eschatology', 'prophethood', 'legal')",
"description": "string",
"related_verse_ids": "array of Verse IDs",
"classical_treatment": "string (how classical tafs■r addresses this theme)",
```

```
"source_id": "string"
}
}
```

5a.12 Legal_Ruling

```
{
"node_type": "Legal_Ruling",
"properties": {
"ruling_id": "string",
"ruling_type": "string (w■jib, mand■b, mub■■, makr■h, ■ar■m)",
"madhab": "string",
"derived_from_verse_ids": "array",
"derivation_method": "string (na■■, ■■hir, mafh■m al-muw■faqa, mafh■m al-mukh■lafa, qiy■s,
etc.)",
"scholarly_consensus_level": "string",
"source_id": "string",
"confidence_score": "float (0–1)"
}
}
```

5a.13 Theological_Position

```
{
"node_type": "Theological_Position",
"properties": {
"position_id": "string",
"aqeedah_school": "string (Ash■ar■ | M■tur■d■ | Athar■ | Mu■tazil■ | Sh■■■ Im■m■ | other)",
"claim_text": "string",
"evidence_verse_ids": "array",
"counter_positions": "array of Theological_Position IDs",
"consensus_status": "string",
"source_id": "string",
"confidence_score": "float (0–1)"
}
}
```

5b. Key Relationship Types (Edge Types)

Each edge carries the provenance schema properties listed in Section 5c.

Edge Type	From Node	To Node	Description
`WORD_IN_VERSE`	Word	Verse	Positional membership
`VERSE_IN_SURAH`	Verse	Surah	Positional membership
`WORD_HAS_ROOT`	Word	Root	Morphological derivation

Edge Type	From Node	To Node	Description
`WORD_HAS_LEMMA`	Word	Root (dictionary head)	Lexical base form
`VERSE_EXPLAINS_VERSE`	Verse	Verse	Tier 1: internal cross-reference; carries `relationship_class` (specifies, generalizes, contextualizes, parallels)
`VERSE_SUPPORTED_BY_HADITH`	Verse	Hadith	Tier 2: Prophetic tafsīr link
`TAFSIR_CLAIM_CITES_EVIDENCE`	Tafsir_Claim	Any node	Source backing for a claim
`TAFSIR_CLAIM_DISPUTES_CLAIM`	Tafsir_Claim	Tafsir_Claim	Documented scholarly disagreement
`TAFSIR_CLAIM_SUPPORTS_CLAIM`	Tafsir_Claim	Tafsir_Claim	Corroborating scholarly opinion
`VERSE_HAS_ASBAB`	Verse	Asbab_al_Nuzul	Occasion of revelation linkage
`WORD_HAS_QIRAA_VARIANT`	Word	Qiraa	Variant recitation affecting the word
`CLAIM_SUPPORTS_RULING`	Tafsir_Claim	Legal_Ruling	Exegetical basis for fiqh derivation
`SCHOLAR_AUTHORED_TAFSIR`	Scholar	Tafsir_Claim	Intellectual ownership
`THEME_CONTAINS_VERSE`	Theme	Verse	Thematic membership
`VERSE_HAS_RHETORICAL_DEVICE`	Verse	Rhetorical_Device	Stylistic annotation
`CLAIM_CONTRADICTS_VERSE`	Tafsir_Claim	Verse	Flags when a claim contradicts another verse's clear text
`CLAIM_ALIGNS_WITH_POSITION`	Tafsir_Claim	Theological_Position	Maps exegesis to ʿaqīdah position

Edge Type	From Node	To Node	Description
VERSE_REFINES_LIMITS	Verse	Verse	Directed chronological edge: a later-revealed verse specifies, narrows, or completes the scope of an earlier general verse. Carries revelation-order metadata. Used for progressive revelation chains (e.g., 5:90 refines 4:43 which refines 2:219 on intoxicants).
VERSE_PRECEDES_IN_REVELATION	Verse	Verse	General temporal-ordering edge: verse A was revealed before verse B. Distinct from VERSE_REFINES_LIMITS (which requires a substantive scope relationship). Carries revelation_period and chronological_order properties.

5c. Provenance Schema

Every node and edge in the graph must carry these metadata properties, without exception:

```
{
"provenance": {
"source_id": "string (unique ID of the source document or dataset)",
"source_type": "enum: classical_text | hadith_collection | lexicon | modern_scholarly |
human_annotation…
"tier": "integer (1–7, evidential tier of this source)",
"confidence_score": "float (0.0–1.0, computed by combination of source reliability and
annotation qualit…
"confidence_rationale": "string (brief human-readable explanation of the score)",
"date_entered": "ISO8601 datetime",
"last_reviewed": "ISO8601 datetime",
"reviewer_id": "string (human reviewer responsible for last validation)",
"automated_flag": "boolean (was this node/edge created by the NLP pipeline?)",
"human_verified": "boolean",
"disputed": "boolean (is there a recorded scholarly dispute about this?)",
"dispute_ref": "array of Tafsir_Claim IDs documenting the dispute"
}
```

```
}
```

Design principle: No claim enters the knowledge graph without a source_id and tier assignment. Automated pipeline outputs begin with `human_verified: false` and must be reviewed before being used in scoring outputs above Low confidence.

5c-1. Retrieve-Don't-Generate (RDG) Principle (v2.1)

Core rule: The system distinguishes between two categories of AI operation:

1. AI-as-Analyst (permitted to compute): Pattern detection, morphological decomposition, semantic embeddings, structural analysis, cross-reference candidate detection, scoring computation, rhetorical device identification, and statistical measurement performed directly on the Qur'anic text or on verified data already in the knowledge graph. These are computational operations on visible inputs — the AI processes data it can inspect, and its outputs are verifiable against those inputs.

2. AI-as-Scholar-Proxy (must retrieve, never generate): Any factual claim about what a human source (scholar, lexicon, hadith collection, tafsīr work) says. These claims must be retrieved from indexed, authenticated source corpora — never generated from an LLM's parametric memory, even when the model is highly confident.

Why this matters: LLMs do not fabricate randomly — they fabricate plausibly. A model can generate "Al-Zamakhsharī notes in al-Kashshāf that the morphological form faʿala here carries an iterative sense" — a sentence that pattern-matches perfectly to real scholarship but may be entirely invented. Retrieval from indexed sources eliminates this failure mode.

RDG Source Authority Table

Every factual claim in the knowledge graph must trace to a retrieval from the following indexed sources. LLM parametric memory is never an acceptable source.

Data Type	Indexed Retrieval Source(s)	Never Generate From
Hadith text and grade	Kutub al-Sitta databases (Sunnah.com, Dorar.net), Shamela, IslamWeb	LLM memory
Lexical definitions	Lane's Lexicon (digitized), Lisān al-ʿArab (OpenITI), al-Mufradāt fī Gharīb al-Qurʾān (al-Rāghib), al-Qāmūs al-Muḥīṭ	LLM memory

Data Type	Indexed Retrieval Source(s)	Never Generate From
Morphological analysis	Quranic Arabic Corpus (corpus.quran.com), CAMeL Tools pipeline output (deterministic, not generative)	LLM inference or paraphrase
Qirāʾāt variants	Authenticated qirāʾāt databases; al-Nashr fī al-Qirāʾāt al-ʿAshr (Ibn al-Jazarī, digitized)	LLM memory
Tafsīr claims	Indexed classical tafsīr corpus (OpenITI: al-Ṭabarī, al-Zamakhsharī, al-Rāzī, Ibn Kathīr, al-Qurṭubī, Ibn ʿĀshūr)	LLM paraphrase of training data
Scholarly positions	Indexed scholarly works with page/section references; modern databases with DOI or stable identifiers	LLM memory of what a scholar "likely" said
Cross-references (Verse → Verse)	Pre-computed similarity index from corpus embeddings + Quranic Arabic Corpus thematic annotations + classical tafsīr cross-reference tables	LLM free-association of "similar" verses
Asbāb al-nuzūl	al-Wāḥidī (digitized), al-Suyūṭī's Lubāb al-Nuqūl, hadith databases for narration-based asbāb	LLM memory
Legal rulings	Indexed fiqh databases; al-Mawsūʿa al-Fiqhiyya (Kuwait Encyclopedia of Islamic Jurisprudence, digitized)	LLM inference
Pre-Islamic attestation	Corpus of pre-Islamic poetry (digitized dīwāns), Lane's citations of jāhilī usage	LLM memory

Clarification — what AI *can* compute without retrieval:

Operation	Why Permitted
Root extraction from Arabic text	Deterministic morphological algorithm (CAMeL Tools), not LLM generation
Semantic embeddings / vector similarity	Computed directly on the text — the math is inspectable

Operation	Why Permitted
Rhetorical device detection	Pattern classifier trained on labeled data — outputs are probabilistic but verifiable
D1–D5 scoring	Formula computation on verified inputs — the formula is transparent
Structural pattern detection (ring composition, chiasm)	Algorithmic analysis on the text — reproducible and auditable
Divergence calculation (TLS – ITS)	Pure arithmetic on scored dimensions

Enforcement mechanism: The Citation Verification Gate (Section 5e, NLP Pipeline) intercepts all Scholar-Proxy outputs before graph ingestion. Any factual claim that cannot be matched to an indexed source is rejected and logged for human review.

5d. Sample Graph Fragment — Verse 2:255 (Āyat al-Kursī)

The following JSON-like representation is a compressed illustration, not a production-ready format. It demonstrates how four interconnected nodes might look with their relationships.

```
{
"nodes": [
{
"id": "verse:2:255",
"node_type": "Verse",
"properties": {
"verse_id": "2:255",
"surah_id": 2,
"aya_number": 255,
"text_arabic": "[redacted] ...",
"structural_position": "Structural center of Sura 2 (ring composition — Farrin analysis)",
"muhkam_or_mutashabih": "Muhkam",
"makki_madani_override": "Madan[redacted]",
"confidence_score": 0.98,
"source_id": "tanzil-net-xml-v1",
"last_reviewed": "2025-01-01",
"reviewer_id": "scholar:classical-review-001"
}
},
{
"id": "word:2:255:4",
"node_type": "Word",
"properties": {
"word_id": "2:255:4",
"surface_form": "[redacted]",
"surface_form_vocalized": "[redacted]",
"lemma": "[redacted]",
```

```
"root": "■■■",
"POS_tag": "noun",
"morphological_features": {
"case": "nominative",
"number": "singular",
"gender": "masculine",
"definiteness": "definite"
},
"semantic_domain_primary": "divine attribute (■ifa dh■tiyya)",
"polysemy_flag": false,
"confidence_score": 0.99,
"source_id": "qac-corpus-annotation",
"last_reviewed": "2025-01-01",
"reviewer_id": "scholar:nlp-review-001"
}
},
{
"id": "root:■■■",
"node_type": "Root",
"properties": {
"root_id": "■■■",
"root_form": "■-■-■",
"root_type": "trilateral",
"semantic_field_primary": "life, living, vitality",
"lisan_alarab_entry": "■■■■■■■ ■■ ■■■■■■... ■■■ ■■■■: ■■■ ■■■ ■■ ■■■ ■■■■■■...",
"lane_lexicon_entry": "Hayy: alive; living... al-■ayy as a divine name: The Ever-Living, the
Self-Su…
"alraghib_entry": "■■■■■■■ ■■■■■ ■■■ ■■■■■: ■■■■■■■: ■■■■■■■■ ■■■■■■■■■■ ■■■■■■
■■■■■■■■■■...",
"quran_occurrence_count": 19,
"confidence_score": 0.99,
"source_id": "lisan-alarab-digitized"
}
},
{
"id": "tafsir_claim:tabari:2:255:hayy",
"node_type": "Tafsir_Claim",
"properties": {
"claim_id": "tabari:2:255:hayy",
"verse_id": "2:255",
"claim_text_arabic": "■■■■■■■: ■■■■■ ■■ ■■■■■■ ■■■ ■■■■■■■ ■■■■■ ■■ ■■■■■",
"claim_text_english": "al-■ayy: The One who does not die; He who endures without decay",
"claim_type": "lexical + theological",
"layer": 3,
"tier": 5,
"source_scholar_id": "scholar:tabari",
"source_work": "J■mi■ al-Bay■n ■an Ta■w■l ■y al-Qur■■n",
"school": "Early Classical Sunni",
"consensus_status": "mashh■r",
"confidence_score": 0.91,
"source_id": "tabari-tafsir-digitized-v2",
"last_reviewed": "2025-01-01",
"reviewer_id": "scholar:classical-review-001"
}
}
```

```
],
"edges": [
{
"from": "word:2:255:4",
"to": "verse:2:255",
"type": "WORD_IN_VERSE",
"properties": {
"position": 4,
"source_id": "tanzil-net-xml-v1",
"tier": 1,
"confidence_score": 0.99
}
},
{
"from": "word:2:255:4",
"to": "root:■■■",
"type": "WORD_HAS_ROOT",
"properties": {
"source_id": "qac-corpus-annotation",
"tier": 6,
"confidence_score": 0.99,
"automated_flag": true,
"human_verified": true
}
},
{
"from": "tafsir_claim:tabari:2:255:hayy",
"to": "verse:2:255",
"type": "TAFSIR_CLAIM_CITES_EVIDENCE",
"properties": {
"relationship_direction": "claim_about",
"source_id": "tabari-tafsir-digitized-v2",
"tier": 5,
"confidence_score": 0.91
}
},
{
"from": "verse:2:255",
"to": "verse:3:2",
"type": "VERSE_EXPLAINS_VERSE",
"properties": {
"relationship_class": "parallels",
"description": "Q3:2 (All■hu l■ il■ha ill■ huwa l-■ayyu l-qayy■m) repeats the same divine
attributes…
"source_id": "tier1-internal-crossref-v1",
"tier": 1,
"confidence_score": 0.97
}
}
]
}
```

5e. Proposed Technology Stack

Graph Database: Neo4j vs. Amazon Neptune

Criterion	Neo4j	Amazon Neptune
Query language	Cypher (expressive, widely known)	openCypher + Gremlin + SPARQL (more flexible, steeper learning curve)
Deployment	Self-hosted or AuraDB (cloud)	AWS-native, fully managed
Performance (scale)	Excellent for ≤500M nodes; requires clustering for larger	Scales horizontally; better for very large graphs
RDF / linked data	Limited native RDF support	Native RDF/SPARQL — better for integration with Semantic Web standards
Developer ecosystem	Larger community; more tutorials	Growing, but smaller
Recommendation for this project	Preferred for early phases — Cypher is more legible for humanities scholars; AuraDB reduces ops overhead	Consider Neptune in Phase 4 if RDF alignment with Qur'anic linked data initiatives is desired

Recommendation: Begin with Neo4j AuraDB in phases 1–3; evaluate migration to Neptune or a hybrid store in Phase 4 based on scale requirements and linked data strategy.

NLP Pipeline (Updated v2.1 — with Citation Verification Gate)

```
Text Input (Arabic)
↓
CAMeL Tools Preprocessing (normalization, tokenization)
↓
CAMeL Tools Morphological Analyzer (root, lemma, POS, morphological features) [AI-as-Analyst:
deterministic]
↓
CAMeLBERT-CA (Classical Arabic) — contextual embeddings [AI-as-Analyst: computed on text]
↓
Custom Fine-tuned Models (trained on Qur'anic Arabic Corpus + tafs■r texts): [AI-as-Analyst]
• Semantic Role Labeling
• Rhetorical Device Detection
• Named Entity Recognition (divine names, places, persons)
• Cross-reference candidate detection
↓
■■■
```

```
■ qir■■■t, asb■b, legal ruling, scholarly position): ■
■ ■
■ 1. Source ID lookup: Does source_id exist in the indexed ■
■ corpus? If no → REJECT + log for human review ■
■ ■
■ 2. Passage match: Does the quoted/cited passage match the ■
■ indexed text (fuzzy match threshold ≥ 0.85)? ■
■ If no → REJECT + log with diff for human review ■
■ ■
■ 3. Claim-source alignment: Does the derived conclusion ■
■ follow from the cited passage? (NLI classifier, ■
■ confidence ≥ 0.70 for auto-pass; below → human review) ■
■ ■
■ Output: PASS / REJECT / HUMAN_REVIEW_REQUIRED ■
■■■
```

Vector Store (for Semantic Search)

Option	Strengths	Weaknesses
Pinecone	Fully managed; excellent performance; simple API	Proprietary; cost at scale
Weaviate	Open source; supports hybrid search (vector + keyword); can store structured objects	More complex to operate
pgvector	Postgres extension; easy if already using Postgres	Less specialized; performance ceiling

Recommendation: Weaviate for its hybrid search capabilities — critical for combining vector-based semantic similarity with structured tier/confidence filters.

API Layer

- FastAPI (Python) — natural fit with the NLP pipeline; async support; auto-generates OpenAPI docs
- GraphQL layer over Neo4j — for flexible front-end queries
- Rate limiting + authentication — required before any public-facing release

Frontend

- React — component-based; large ecosystem
- D3.js — force-directed graph visualization for the knowledge graph view
- Tailwind CSS — utility-first styling
- Arabic text rendering: Use a Unicode-compliant Arabic font stack (Amiri, Noto Naskh Arabic) with RTL layout enforcement throughout

---

5f. Visual Layer Tiering (v2.0)

The knowledge graph UI implements a two-layer visual architecture:

Primary Layer (High Contrast):

- All Qurʾānic structural nodes: Surah, Verse, Word, Root, Qirāʾa, Rhetorical_Device, Theme
- All Qurʾān-internal edges: WORD_IN_VERSE, VERSE_IN_SURAH, WORD_HAS_ROOT, VERSE_EXPLAINS_VERSE, VERSE_REFINES_LIMITS, VERSE_PRECEDES_IN_REVELATION, VERSE_HAS_RHETORICAL_DEVICE, THEME_CONTAINS_VERSE, WORD_HAS_QIRAA_VARIANT
- Rendered in primary teal (#01696F) and full-contrast text (#28251D)

Secondary Layer (Muted):

- All tradition and scholarly nodes: Scholar, Hadith, Asbab_al_Nuzul, Legal_Ruling, Theological_Position, Tafsir_Claim
- All tradition-linked edges: VERSE_SUPPORTED_BY_HADITH, TAFSIR_CLAIM_CITES_EVIDENCE, TAFSIR_CLAIM_DISPUTES_CLAIM, SCHOLAR_AUTHORED_TAFSIR, CLAIM_SUPPORTS_RULING, CLAIM_ALIGNS_WITH_POSITION, VERSE_HAS_ASBAB
- Rendered in muted tones (#7A7974) with a clear "Layer 2/3" label

Design principle: The graph defaults to showing the Primary Layer. The Secondary Layer is toggle-able. When both are visible, the visual hierarchy makes clear that the Qurʾānic structure is the backbone and the scholarly layer is annotation.

Section 6: Interpretation Scoring Design

6a. Five Scoring Dimensions

Dimension 1 — Lexical Fidelity (0–100)

What it measures: Does the submitted interpretation use the Arabic words in ways consistent with their documented semantic range in the Qur'anic and Classical Arabic corpus?

Sub-checks:

- Is the root correctly identified?
- Is the assigned sense present in Lane's Lexicon and/or Lisān al-ʿArab for this root?
- Is the assigned sense attested in Qur'anic usage elsewhere (internal frequency analysis)?
- Does the assigned sense respect the morphological form (e.g., a masdar form cannot bear the same sense as an active participle)?
- Does the interpretation acknowledge (or unjustifiably exclude) the qirāʾāt variant if one exists for this word?
- Does the interpretation honor the grammatical role of the word in the sentence?

Scoring calibration:

- 90–100: Interpretation uses the word in its most well-documented Qur'anic sense; acknowledges polysemy where relevant
- 70–89: Interpretation uses a documented but less frequent sense; no morphological violations
- 50–69: Interpretation uses a sense that is attested in classical Arabic but rare or non-standard in Qur'anic usage
- 30–49: Interpretation requires a sense that is a stretch of the documented semantic range; requires additional justification
- 0–29: Interpretation requires a sense not found in classical Arabic lexicography, or directly violates morphological constraints

---

Dimension 2a — Thematic Context (0–100)

What it measures: Does the interpretation fit the immediate textual environment — the surrounding verses, the surah's thematic arc, and Qur'anic intertextuality as observable from the muṣḥaf arrangement?

Sub-checks:

- Is it consistent with the preceding and following verses in the passage?
- Does it fit the Makkī/Madanī register (addressee patterns, vocabulary, characteristic concerns)?
- Does it account for established thematic clusters?
- Is it consistent with the surah's structural features (ring composition, thematic progression) where documented?

Scoring calibration:

- 90–100: Fits all documented contextual dimensions; actively strengthened by context
- 70–89: Consistent with context; does not contradict any contextual element
- 50–69: Partially fits context; requires selective reading of contextual elements
- 30–49: Creates tension with one or more documented contextual elements
- 0–29: Directly contradicts the documented occasion of revelation, or is inconsistent with the passage's clear textual environment

D2a Boundary Definition (v2.0): Pass 1's D2a uses Makkī/Madanī classification because it is a textual-structural property — the Arabic register, addressee patterns (yā ayyuhā alladhīna āmanū vs. yā ayyuhā al-nās), and thematic vocabulary differ measurably between the two periods. These are observable linguistic features, not external tradition. What D2a strictly excludes: specific asbāb al-nuzūl narrations (e.g., "this verse was revealed about ʿUbāda ibn al-Ṣāmit's alliance"), named historical incidents, and any contextual claim that depends on an isnād chain. Those belong to Pass 2.

Dimension 2b — Temporal Context (0–100)

What it measures: Does the interpretation account for the verse's position in the revelation sequence? Does it treat a verse as final or standalone when a later revelation qualifies, refines, or supersedes it?

Sub-checks:

- Does the interpretation acknowledge the verse's revelation period (early Makkī through late Madanī)?
- If the verse is part of a progressive revelation chain (e.g., intoxicant prohibition: 2:219 → 4:43 → 5:90), does the interpretation account for which stage it belongs to?
- Does the interpretation ignore a later verse that modifies this ruling (i.e., does it cite a mansūkh verse without acknowledging the nāsikh)?
- Does the verse's meaning depend on when it was revealed relative to other verses on the same topic?

Scoring calibration:

- 90–100: Interpretation fully accounts for chronological position; acknowledges progressive revelation chains where applicable
- 70–89: Interpretation is consistent with chronological context; no temporal errors
- 50–69: Interpretation partially accounts for chronology; some temporal dimension overlooked
- 30–49: Interpretation ignores a relevant later verse or applies a general ruling without acknowledging its subsequent specification
- 0–29: Interpretation treats a superseded verse as operative, or ignores the revelation sequence in a way that changes the verse's meaning

---

Dimension 3 — Transmitted Support (0–100)

What it measures: Is the interpretation supported by any Tier 1–4 evidence — internal Qur'anic cross-reference, authenticated Prophetic hadith, Companion reports, or early Successor reports?

Sub-checks:

- Does any Tier 1–2 source support this interpretation? (High positive impact)
- Is it consistent with the dominant Companion view?
- Is it found in the tābiʿūn literature?
- If no Tier 1–4 support exists, has any classical scholar noted this absence?
- If it contradicts a Tier 1–2 source, that is a hard negative score cap (ceiling of 40)

Scoring calibration:

- 85–100: Directly supported by ṣaḥīḥ Prophetic tafsīr or clear Qur'an-by-Qur'an evidence
- 65–84: Supported by dominant Companion view or multiple tābiʿūn reports
- 45–64: Supported by some tābiʿūn reports; no direct Companion attestation
- 25–44: No Tier 2–4 support; interpretation is consistent with (but not attested in) the transmitted tradition
- 0–24: Interpretation contradicts one or more Tier 1–4 sources

Hard cap rule: If a user interpretation directly contradicts a ṣaḥīḥ Prophetic hadith that provides specific tafsīr of this verse, Dimension 3 is capped at 20 regardless of other factors, and a mandatory FLAG is issued.

Dimension 4 — Scholarly Precedent (0–100)

What it measures: Has this interpretation or a close analog appeared in classical or respected modern tafsīr? Is it part of the documented scholarly discourse or a novel claim?

Sub-checks:

- Is the interpretation found in any Tier 5 source (al-Ṭabarī, Ibn Kathīr, al-Qurṭubī, al-Baghawī, al-Zamakhsharī, al-Rāzī, Ibn ʿĀshūr)?
- How many distinct schools have endorsed this interpretation?
- Is it a minority (marjūḥ) or majority (mashhūr/rājiḥ) view in the classical tradition?
- If the interpretation is modern, does it cite its reasoning in relation to the classical tradition?
- Is it a known innovation (bidʿa) rejected explicitly by classical scholars?

Scoring calibration:

- 85–100: Widespread classical consensus across multiple schools; appears in 3+ major tafsīr works
- 65–84: Present in at least 2 major tafsīr works; acknowledged as a valid position even if not dominant
- 45–64: Found in one classical tafsīr or multiple modern scholarly works
- 25–44: Not found in classical sources; found only in contemporary discussions
- 0–24: Explicitly rejected or contradicted in classical tafsīr literature

---

Dimension 5 — Internal Consistency (0–100)

What it measures: Does the interpretation create contradictions with other clear Qur'anic verses, or with established readings of the same verse that are considered settled?

Sub-checks:

- Does the interpretation require that another clear verse be read differently or be contradicted?
- Does it create logical or theological inconsistencies within the Qur'anic framework?
- Does it require a special pleading that a muhkam (clear) verse is actually mutashābih (ambiguous) without scholarly precedent for that reclassification?
- Does it require ignoring an abrogating verse (nāsikh) in favor of an abrogated one (mansūkh)?
- Does it create doctrinal inconsistency with verses considered ijmāʿ-level settled?

Scoring calibration:

- 90–100: Internally consistent; actively coherent with related verses
- 70–89: No direct contradiction; some minor tension with related verses that can be resolved
- 50–69: Creates tension with at least one related verse that requires additional explanation
- 30–49: Requires non-standard re-reading of another clear verse to be sustained
- 0–29: Directly contradicts a muhkam (clear, settled) verse

6b. Two-Pass Scoring Architecture (v2.0)

The system generates two independent scores for every submitted interpretation:

Pass 1: Textual Logic Score (TLS)

Objective: Determine what the Qur'anic text establishes on its own, treated as a closed-loop system.

Formula:

```
TLS = (D1 × 0.35) + (D2a × 0.15) + (D2b × 0.15) + (D5 × 0.35)
```

Inputs: Lexical Fidelity (D1), Thematic Context (D2a), Temporal Context (D2b), Internal Consistency (D5).

Excluded: All external hadith reports, scholarly opinions, and asbāb al-nuzūl narrations.

Pass 1 Constraint: Pass 1 treats the Qur'an as a self-contained text. It evaluates whether an interpretation is internally coherent with the Qur'anic corpus. It does not claim to be a complete analysis — see the TREC Flag below for cases where the text itself signals that external clarification is part of the design.

Pass 2: Integrated Tradition Score (ITS)

Objective: Measure how the interpretation aligns with the historical and prophetic application of the text.

Formula:

```
ITS = (D1 × 0.20) + (D2a × 0.10) + (D2b × 0.10) + (D3 × 0.30) + (D4 × 0.20) + (D5 × 0.10)
```

Inputs: All six dimensions — D1, D2a, D2b, D3 (Transmitted Support), D4 (Scholarly Precedent), D5 (Internal Consistency).

Pass 2 Constraint: Pass 2 is formally subordinated. It cannot override a Pass 1 finding without triggering a Structural Divergence Warning. The tradition is available but not dominant.

Structural Divergence Warning

When $|TLS - ITS| \geq 25$ points, the system issues a Structural Divergence Warning — a mandatory alert that highlights where tradition significantly departs from or adds to the text.

Divergence Scenario Mapping

Scenario	Color Band	User-Facing Label
TLS ≥ 70, ITS ≥ 70	Green	Strong alignment — textually sound and tradition-confirmed
TLS ≥ 70, ITS < 45	Yellow + Warning	Textually coherent but departs from transmitted interpretation
TLS < 45, ITS ≥ 70	Orange + Warning	Tradition-dependent — the text alone does not establish this reading
TLS < 30, ITS < 30	Red	Unsupported / contradicted by both text and tradition
TLS 45–69, ITS 45–69	Gray (Ambiguous)	Partial support from both; genuine interpretive gray zone

TREC Flag: Text Requests External Clarification

A mandatory flag triggered when Pass 1 encounters a verse in a class the Qur'an itself marks as requiring the Messenger's explanation. Triggering verses include:

- 16:44 — "We sent down to you the Reminder so that you may make clear to people what has been sent down to them."
- 59:7 — "Whatever the Messenger gives you, take it."
- 4:65 — "They will not truly believe until they make you judge in their disputes."

When a TREC flag fires, the user sees: "The Qur'an's own internal logic directs to the Prophetic tradition for this category of verse. Pass 1 (Textual Logic) is not claiming to be a complete analysis; the text itself requests external clarification."

The TREC flag is a transparency mechanism, not a score override. It does not change the TLS number; it annotates the output with an intellectual honesty marker.

6c. Confidence Score Formula

The Confidence Score (CS) is separate from and orthogonal to the Alignment Score. It answers: How certain is the framework itself in its assessment? A low Confidence Score means the framework lacks sufficient evidence to evaluate the interpretation reliably — not that the interpretation is wrong.

$$
\text{CS} = \alpha \times C_{\text{source}} + \beta \times C_{\text{coverage}} + \gamma \times C_{\text{agreement}}
$$

\]

Where:

- C_source = source quality score (0–1): derived from the average tier of evidence available for this verse; verses with strong Tier 1–2 evidence score high
- C_coverage = coverage score (0–1): what proportion of the relevant evidence has been retrieved and processed?
- C_agreement = inter-annotator agreement score (0–1): if multiple human reviewers or model runs independently assessed this claim, how much did they agree? (Using Krippendorff's α)

Proposed weights: α = 0.40, β = 0.30, γ = 0.30

v2.0 note: The confidence score applies independently to both TLS and ITS. A high-confidence TLS with a low-confidence ITS means the text is clear but the tradition is thin. A low-confidence TLS with a high-confidence ITS means the text is ambiguous but tradition provides a strong reading.

Worked example for a well-covered verse (e.g., 2:255):

- C_source = 0.90 (Tier 1–3 evidence richly available)
- C_coverage = 0.85 (most major tafsīr works digitized and indexed)
- C_agreement = 0.80 (high scholar agreement on core meaning)
- CS = (0.40 × 0.90) + (0.30 × 0.85) + (0.30 × 0.80) = 0.36 + 0.255 + 0.24 = 0.855 → 85.5 / 100

Worked example for a less-covered verse (e.g., some cryptic muqaṭṭaʿāt):

- C_source = 0.40 (limited Tier 1–3 evidence)
- C_coverage = 0.55 (some classical sources digitized)
- C_agreement = 0.35 (significant scholarly disagreement)
- CS = (0.40 × 0.40) + (0.30 × 0.55) + (0.30 × 0.35) = 0.16 + 0.165 + 0.105 = 0.430 → 43.0 / 100

6d. Classification Bands (v2.0)

The two-pass architecture replaces the single four-band CAS system with the Divergence Scenario Mapping from Section 6b. For reference:

Scenario	Color Band	User-Facing Label
TLS ≥ 70, ITS ≥ 70	Green	Strong alignment — textually sound and tradition-confirmed
TLS ≥ 70, ITS < 45	Yellow + Warning	Textually coherent but departs from transmitted interpretation

Scenario	Color Band	User-Facing Label
TLS < 45, ITS ≥ 70	Orange + Warning	Tradition-dependent — the text alone does not establish this reading
TLS < 30, ITS < 30	Red	Unsupported / contradicted by both text and tradition
TLS 45–69, ITS 45–69	Gray (Ambiguous)	Partial support from both; genuine interpretive gray zone

Display rule: If TLS ≥ 70 but CS < 70 (the framework is uncertain even about a high score), annotate the output with a note: "Score appears high but evidence coverage is limited for this verse — treat with additional caution."

6e. Plain-Language Misalignment Explanations

For each scoring dimension, the system generates one or more plain-language flags when the dimension score falls below a threshold (60). These are written for a non-specialist reader.

D1 Flag Template (Lexical Fidelity):

> "The word you interpreted as [USER_MEANING] is [ARABIC_WORD] (root: [ROOT]). In classical Arabic — across [N] occurrences in the Qur'an — this word is most commonly used to mean [PRIMARY_SENSE]. The sense you assigned, [USER_MEANING], [IS / IS NOT] found in classical Arabic dictionaries, but [IS RARE / HAS NOT BEEN ATTESTED] in Qur'anic usage. [IF QIRAA RELEVANT: There is also a recitation variant for this word that [CHANGES / DOES NOT CHANGE] its meaning.] The system has scored your lexical alignment at [D1_SCORE]/100."

Example D1 flag:

> "The word you translated as 'light' is (nūr, root: - -). In this surah specifically, this word is used 5 other times, and in all 5 cases major classical scholars including al-Ṭabarī and Ibn Kathīr interpret it as 'divine guidance' rather than physical light. Your reading of it as literal light is a documented sense of the root, but it is the secondary meaning in this Qur'anic context. Your lexical fidelity score is 62/100."

D2 Flag Template (Contextual Coherence):

> "This verse was revealed [OCCASION OF REVELATION if known — source cited]. Reading it as [USER_INTERPRETATION] creates a tension with this context because [SPECIFIC REASON]. Additionally, the [preceding / following] verse says [RELEVANT ADJACENT TEXT], which

> [supports / complicates] your reading."

D3 Flag Template (Transmitted Support):

> "Your interpretation does not appear to be attested in any reports from the Prophet's Companions or early successors. [IF CONTRADICTS: There is a [ṣaḥīḥ / ḥasan]-graded hadith in which the Prophet explained this verse differently — specifically: [BRIEF HADITH SUMMARY] (source: [KITAB, NUMBER]).] Classical scholars who discuss this passage — including al-Ṭabarī, Ibn Kathīr, and [THIRD SCHOLAR] — do not support your reading. This significantly reduces your transmitted support score to [D3_SCORE]/100."

D4 Flag Template (Scholarly Precedent):

> "Your interpretation does not appear to have been advanced in the major classical tafsīr works. [IF MODERN ANALOGUE: A somewhat similar reading was proposed by [MODERN SCHOLAR, DATE], but it has not been widely adopted.] The dominant classical interpretation of this passage — held by al-Ṭabarī, Ibn Kathīr, and al-Qurṭubī among others — is [CLASSICAL_INTERPRETATION]."

D5 Flag Template (Internal Consistency):

> "Your interpretation creates a tension with Qur'an [CROSS_REFERENCE_VERSE], which states [VERSE TEXT]. Most classical scholars read these two verses as consistent, which requires [BRIEF RECONCILIATION LOGIC]. If your interpretation is correct, it would require reading [CROSS_REFERENCE_VERSE] differently from how the tradition has understood it."

6f. Nearest Better-Framed Interpretation (NBFI) Algorithm

NBFI v2.0: Intra-Textual Priority

The Nearest Better-Framed Interpretation algorithm now operates in two passes:

NBFI Pass 1 (Default View): Surfaces only Qur'anic parallels — Tier 1 evidence. Uses the `VERSE_EXPLAINS_VERSE`, `VERSE_REFINES_LIMITS`, and `VERSE_PRECEDES_IN_REVELATION` edges to find intra-textual alternatives. Example: if a user submits a political reading of khalīfa in 2:30, NBFI Pass 1 retrieves 38:26 (Dāwūd's stewardship) as a linguistic anchor showing the term implies "accountability" rather than "political office."

NBFI Pass 2 (Toggle View): Adds transmitted scholarly alternatives from Tier 2–5 sources. Available as a user-activated toggle.

Design principle: The user sees what the Qur'an itself says first. Scholarly alternatives are one click away, never hidden, but never front-running the text.

Purpose: When a user's interpretation scores in the Orange or Red bands, the system should surface the closest well-evidenced interpretation — the "upgrade path" — rather than simply telling the user they are wrong.

Algorithm:

```
function find_NBFI(user_claim, verse_id):
1. RETRIEVE all Tafsir_Claim nodes where:
- verse_id matches
- CAS_pre_computed >= 60
- tier <= 5
2. EMBED user_claim using CAMeLBERT-CA → user_vector
3. FOR each candidate_claim in retrieved set:
EMBED candidate_claim.text → candidate_vector
cosine_similarity = cos_sim(user_vector, candidate_vector)
4. RANK by:
score = (0.60 × cosine_similarity) + (0.25 × normalized_CAS) + (0.15 × tier_bonus)
where tier_bonus = (8 - candidate.tier) / 7 [lower tier = higher bonus]
5. RETURN top 3 candidates with:
- claim_text (in English and Arabic)
- CAS of the candidate
- tier and source
- brief explanation of how it differs from the user's claim
- the key dimension on which it scores higher
```

Example output block:

> You submitted: "Khalīfa here means political caliph — a ruler with divine authority to govern all Muslims." Your score: CAS = 28/100 (Red — Unsupported)
>
> Nearest well-supported alternatives:
>
> 1. "Khalīfa means a steward of God's creation on earth (vicegerent) — a role given to humanity collectively to manage and be accountable for the earth." — CAS 74/100 (Yellow), supported by dominant classical reading (al-Ṭabarī, Ibn Kathīr, al-Qurṭubī). How it differs from yours: It de-emphasizes the political dimension and centers accountability and stewardship; "all Muslims" is not in the text.
>

2. "Khalīfa refers to the human capacity to succeed one another (successors/deputies) on earth, implementing divine guidance generation by generation." — CAS 68/100 (Yellow), supported by some classical readings and Ibn ʿĀshūr's modern synthesis. How it differs: The succession sense is present but understood generationally rather than as a title.

>

3. "Khalīfa is Ādam specifically, as a successor to those who preceded him, with the angels' question indicating divine foreknowledge of human propensity for corruption." — CAS 80/100 (Green), directly supported by Ibn ʿAbbās's report and al-Ṭabarī's synthesis. How it differs from yours: The referent is Ādam specifically, not a political office; the "divine authority" framing is not in the classical sources.

Section 7: AI Agent Timeline Estimates

The following table assumes a dedicated AI agent (GPT-4 class or equivalent, with tool use) operating with structured prompting and access to digitized classical sources. Human review times assume 2 senior Islamic scholars and 1 NLP specialist working in parallel.

#	Deliverable	AI Agent Time	Human Review Time	Milestone Gate
1	Research blueprint — full methodology document, evidence hierarchy, ontology sketch	2–4 days	2 weeks (scholar validation)	Approved methodology document; sign-off from at least 1 senior tafsīr scholar
2	Ontology draft — complete knowledge graph schema with all node and edge types, provenance model, constraints	3–5 days	2–3 weeks	Schema review by graph architect + scholar; Neo4j constraint file passing validation
3	Corpus acquisition and preprocessing — Tanzil.net XML (full Qur'an), OpenITI classical tafsīr texts (al-Ṭabarī, Ibn Kathīr, al-Qurṭubī in Arabic), Quranic Arabic Corpus annotation	1–2 weeks (retrieval and format standardization)	3–4 weeks (rights review, quality audit)	Corpus inventory document; 100% Qur'anic text coverage; ≥3 major tafsīr works in clean structured format
4	Benchmark dataset — 100 test verses with human-annotated ground-truth interpretations, 5 scoring dimensions, inter-annotator agreement ≥ 0.70 (Krippendorff's α)	2–3 weeks (candidate selection, initial scoring)	6–8 weeks (scholar annotation, adjudication)	IAA ≥ 0.70; documented annotation guidelines; locked benchmark set

#	Deliverable	AI Agent Time	Human Review Time	Milestone Gate
4a	Hallucination detection benchmark (v2.1) — Adversarial test set of 200+ known-false but plausible-sounding scholarly claims across all RDG categories: fabricated hadith references (50), invented lexical definitions (40), false tafsīr attributions (40), non-existent qirāʾāt variants (20), fabricated cross-references (25), false scholarly positions (25). Each item is designed to be fluent, domain-appropriate, and difficult to detect without source verification.	1–2 weeks (AI-generated adversarial examples + human curation)	3–4 weeks (scholar verification that all "false" items are genuinely false and all "true" items are genuinely true)	CVG recall ≥ 95% (no more than 5% of fabricated claims pass); CVG precision ≥ 90% (no more than 10% of real claims falsely rejected); documented error analysis for all failures
5	Pilot scoring system — first 10 surahs (al-Fātiḥa + Juzʾ ʿAmma + 3 major medium surahs), rule-based + retrieval, no ML fine-tuning yet	3–4 weeks	4–6 weeks (scholar QA on 500+ test scores)	System scores ≥ 75% agreement with human benchmark on pilot set; documented error analysis
6	NLP model fine-tuning — CAMeLBERT-CA fine-tuned on tafsīr corpus for semantic similarity, rhetorical device detection, cross-reference candidate detection	2–4 weeks (training runs)	3–4 weeks (evaluation, error analysis, edge case review)	Model achieves ≥ 80% precision/recall on held-out test set for each subtask; documented ablation results

#	Deliverable	AI Agent Time	Human Review Time	Milestone Gate
7	v1 production framework — all 114 surahs covered; scoring API deployed; React + D3 frontend; full knowledge graph populated	6–10 weeks (integration, testing, deployment)	8–12 weeks (comprehensive scholar review, user testing with 20+ diverse users)	All 114 surahs indexed; API response time < 3 seconds; ≥ 85% scholar satisfaction on QA sample; documented governance policy

Total estimated timeline (AI + human in parallel): 18–24 months (see Section 9 for phased breakdown)

Key bottleneck: Human review time consistently dominates. The AI can produce fast drafts; the limiting factor is scholar availability and the depth of review required for a theologically sensitive domain.

Section 8: Risk Register

#	Risk	Category	Probability	Impact	Mitigation
1	Sectarian bias baked into training data — dominant Sunni Ḥanbalī/Salafī tafsīr sources o verrepresented in digitized corpora; Shīʿī, Ibāḍī, Muʿtazilī, and South/Sout heast Asian scholarly traditions under represented	Scholarly / Governance	High	High	Explicit corpus diversity audit before training; mandatory representation of major schools at Tier 5; governance board with cross-sectarian membership; flag all outputs with "school representation coverage" metadata
2	Overclaiming certainty on genuinely disputed verses — the scoring system produces a single CAS number that users may treat as authoritative; ikhtilāf (legitimate disagreement) gets flattened	Philosophical / Governance	High	High	Mandatory display of Confidence Score alongside CAS; explicit ikhtilāf flag for verses with documented scholarly disagreement; "multiple valid readings" output mode; UX design that foregrounds uncertainty

#	Risk	Category	Probability	Impact	Mitigation
3	Model hallucination of scholarly citations (generalized) — LLM pipeline generates plausible-sounding but non-existent references to hadith, lexicons, tafsīr works, scholarly positions, or qirāʾāt variants to support a score. This is not limited to hadith: an LLM can fabricate lexical definitions, invent tafsīr attributions, or assert false cross-references with equal fluency.	Technical / Scholarly	High	Very High	RDG Principle (Section 5c-1): System must never generate any factual claim about external sources from LLM parametric memory. All such claims must be retrieved from indexed corpora per the RDG Source Authority Table. Citation Verification Gate (Section 5e): Three-step automated verification (source ID lookup, passage match, claim-source alignment) intercepts all Scholar-Proxy outputs before graph ingestion. Items that fail any step are rejected or routed to human review. Human review of all Tier 2 citations before display.

#	Risk	Category	Probability	Impact	Mitigation
3a	Lexical hallucination in Pass 1 — LLM asserts that a classical lexicon (Lane, Lisān al-ʿArab, al-Mufradāt) supports a particular word sense when the indexed entry does not. Especially dangerous because Pass 1 (TLS) is designed to operate without tradition-based correction, so a false lexical claim propagates unchecked into D1 scoring.	Technical	High	High	RDG Source Authority Table mandates retrieval from digitized lexicons only. Citation Verification Gate Step 2 (passage match) catches fabricated definitions. D1 scoring must not proceed on any lexical claim that has not passed the CVG.

#	Risk	Category	Probability	Impact	Mitigation
3b	Plausible scholarly attribution — LLM generates a claim like "al-Zamakhsharī argues in al-Kashshāf that..." using text that pattern-matches to real scholarship but is entirely invented. This is the highest-risk hallucination category because domain experts may not catch it without checking the primary source.	Technical / Scholarly	Medium-High	Very High	Citation Verification Gate Step 2 (passage match against indexed OpenITI corpus) is the primary defense. Step 3 (NLI claim-source alignment) catches cases where the source is real but the attributed conclusion is not. Mandatory `source_id` + `page_ref` on all Tafsir_Claim nodes.
4	Missing qirāʾāt variants that change meaning — the system analyzes the Ḥafṣ ʿan ʿĀṣim reading by default; other canonical recitations (Warsh, Qālūn, Ibn Kathīr) may produce different morphological forms	Technical / Scholarly	Medium	High	Qirāʾāt table must be a first-class schema entity; all 10 canonical qirāʾāt (and the 4 shādhdh ones if relevant) indexed; scoring system must check for qirāʾā variants before finalizing D1 score; flag all verses with meaning-altering variants

#	Risk	Category	Probability	Impact	Mitigation
5	Conflating rhetorical and legal readings — a verse using figurative language (istiʿāra, kināya) is scored as a legal statement, or vice versa; the system does not distinguish between what a verse means rhetorically and what legal scholars derive from it	Scholarly	Medium	High	Rhetorical_Device nodes are first-class; claim_type field on Tafsir_Claim distinguishes rhetorical from legal; explicit warning when a user applies a legal interpretation to a verse classified as primarily rhetorical in classical tafsīr
6	No single authoritative ground truth for genuinely disputed cases — the system requires benchmark "correct answers" for training/evaluation, but the Islamic scholarly tradition deliberately preserves multiple valid positions	Philosophical	High	Medium	Benchmark dataset uses "acceptable range" annotation rather than single correct answer; use NDCG-style ranking metrics rather than accuracy; train evaluators to score plausibility rather than correctness; document this limitation prominently in all published materials

#	Risk	Category	Probability	Impact	Mitigation
7	Political and religious backlash — the system is perceived as claiming AI authority over Qur'anic interpretation; hostile framing by political/religious actors regardless of actual system design	Governance / Political	High	Very High	Clear public framing as an " evidence-ranking tool, not a fatwa machine" from launch; advisory board of respected scholars from multiple traditions; prominent disclaimers; opt-in review process where scholars can dispute outputs; open governance model
8	Scholar refusal to validate AI-assisted tafsīr — senior scholars decline to review or endorse outputs, depriving the system of the credibility it requires	Governance / Scholarly	Medium-High	High	Early engagement with scholars as co-designers, not reviewers; framing as a research tool; early academic publications establishing methodology; scholars credited prominently; avoid framing as "AI tafsīr" — frame as "evidence retrieval and ranking"

#	Risk	Category	Probability	Impact	Mitigation
9	Corpus copyright issues for classical texts — digitized classical tafsīr works may have copyright restrictions depending on edition; modern translations and commentaries (Ibn ʿĀshūr's English translation, The Study Quran) are under copyright	Legal	Medium	Medium	Prioritize OpenITI corpus (open access classical texts); use classical Arabic primary sources directly rather than modern translations; obtain explicit permissions for any modern translations used; include only metadata pointers to commercial works, not full text
10	Isnād reliability automation failure — automated hadith grading using NLP produces incorrect reliability assessments; ḍaʿīf narrations get coded as ṣaḥīḥ	Technical / Scholarly	High	Very High	Never auto-grade isnāds; use only pre-existing human scholarly grades from established hadith databases (Dorar.net, Islamweb, Sunnah.com); flag all Tier 2 sources with their human-attributed grade; manual expert review of all hadith-verse linkages

#	Risk	Category	Probability	Impact	Mitigation
11	NLP pipeline failure on archaic morphology — CAMeLBERT and CAMeL Tools trained primarily on Modern Standard Arabic; Qur'anic Classical Arabic has constructions not well-represented in training data	Technical	Medium-High	Medium	Use CAMeLBERT-CA (classical Arabic variant specifically); supplement with Quranic Arabic Corpus annotations as training signal; maintain human fallback for morphologically ambiguous forms; flag all cases where model confidence is low
12	Isrāʾīliyyāt contamination in classical sources — narratives of Jewish or Christian origin that entered some tafsīr works are indexed alongside authentic transmitted interpretation	Scholarly	Medium	Medium	Explicit Isrāʾīliyyāt flag on Tafsir_Claim nodes; tag sources known for high Isrāʾīliyyāt content (some reports attributed to Kaʿb al-Aḥbār, Wahb ibn Munabbih); reduce tier weighting for unverified Isrāʾīliyyāt material

#	Risk	Category	Probability	Impact	Mitigation
13	User gaming of the scoring system — sophisticated users learn to manipulate claim phrasing to score higher without substantive alignment	Technical / Governance	Low-Medium	Low	Log all submissions; monitor for adversarial patterns; rate-limit scoring API; require users to cite their own sources; design for educational transparency so gaming is visible
14	Asbāb al-nuzūl over-restriction — system flags interpretations as contextually inconsistent because they apply a verse beyond its specific occasion of revelation, contrary to the classical rule that the lesson is the generality of the wording (al-ʿibra bi-ʿumūm al-lafẓ)	Philosophical / Technical	Medium	Medium	Encode the ʿumūm al-lafẓ rule explicitly in D2 scoring logic; distinguish between "contextually inconsistent with the occasion" (legitimate flag) and "extending beyond the occasion" (not a flag by default)

#	Risk	Category	Probability	Impact	Mitigation
15	Conflation of historical and normative authority — users interpret a high Alignment Score as meaning the interpretation is Islamically binding, or that AI assessment of an interpretation has replaced scholarly authority	Governance	Medium	High	UX copy must explicitly state that a high score means "well-evidenced by the scholarly tradition" not "correct" or "binding"; legal disclaimer; onboarding flow that educates users on what the score does and does not mean
Pass 1 false self-sufficiency	Philosophical	Medium	High	The TREC flag and D2b temporal check counteract this; the system never claims Pass 1 is complete	
Divergence threshold gaming	Technical	Low	Medium	Monitor for users reverse-engineering interpretation phrasing to avoid the ≥25pt divergence trigger; adjust threshold periodically	

#	Risk	Category	Probability	Impact	Mitigation
Chronological ordering uncertainty	Scholarly	High	Medium	Many verse-level chronological assignments are disputed; the revelation_period_confidence property (High/Medium/Low) tracks this; flag "Low" confidence chronology explicitly	

Section 9: Phased Build Plan

Phase 1: Foundation (Months 1–3)

Objective: Establish scholarly, methodological, and technical foundations before writing a single line of production code.

Key Activities:

1. Recruit core team: Qur'anic Methodology Lead, Classical Tafsīr Scholar, Arabic NLP Scientist, Knowledge Graph Architect
2. Finalize and publish methodology document (this document, refined)
3. Conduct evidence hierarchy workshops with scholar team; document all edge cases and disagreements
4. Inventory available digitized sources: Tanzil.net XML, OpenITI corpus, Quranic Arabic Corpus
5. Develop annotation guidelines for benchmark dataset
6. Establish governance framework: advisory board composition, dispute resolution process, sectarian bias policy
7. Technical feasibility study: Neo4j vs. Neptune, CAMeL Tools pipeline evaluation, vector store benchmarking
8. Implement two-pass scoring architecture spec
9. Define D2a/D2b boundary rules for all 114 surahs
10. Establish revelation_period assignments for all surahs using al-Suyūṭī, Nöldeke, and consensus lists

Deliverables:

- Signed-off methodology document
- Annotation guidelines v1.0
- Corpus inventory and rights clearance report
- Governance charter v1.0
- Technical architecture decision record (ADR)

Milestone Gate: Scholar team consensus on methodology; governance charter approved; technology stack selected

Team Required: Qur'anic Methodology Lead (100%), Tafsīr Scholar (50%), Arabic NLP Scientist (50%), Knowledge Graph Architect (50%), Governance Reviewer (25%)

Phase 2: Core Build (Months 4–6)

Objective: Build the knowledge graph schema, NLP preprocessing pipeline, and initial evidence database for the first set of surahs.

Key Activities:

1. Implement Neo4j graph schema with all node types, edge types, and provenance model from Section 5
2. Ingest Tanzil.net XML → graph (all 114 surahs, word-level annotation)
3. Run CAMeL Tools morphological pipeline on full Qur'anic text; human review of flagged cases
4. Ingest Quranic Arabic Corpus POS and morphological annotations into graph
5. Begin al-Ṭabarī, Ibn Kathīr, al-Qurṭubī structured ingestion (Arabic text, verse-by-verse claim extraction)
6. Build scoring API prototype (rule-based, no ML fine-tuning yet) for 5 pilot surahs
7. Annotate 50-verse benchmark sample; run first IAA analysis
8. Build VERSE_REFINES_LIMITS and VERSE_PRECEDES_IN_REVELATION edges for progressive revelation chains
9. Implement TREC flag triggering logic
10. Implement Structural Divergence Warning with ≥25pt threshold
11. Implement NBFI v2.0 two-pass retrieval

Deliverables:

- Neo4j graph with full Qur'anic text and morphological annotations
- Tier 5 ingestion pipeline for 3 major tafsīr works
- Scoring API v0.1 (rule-based)
- Benchmark sample annotated by at least 2 scholars; IAA report

Milestone Gate: Graph populated with 100% of Qur'anic text; scoring API functional for pilot surahs; IAA ≥ 0.65 on 50-verse sample

Team Required: Knowledge Graph Architect (100%), Arabic NLP Scientist (100%), Tafsīr Scholar (100%), Methodology Lead (50%), ʿUlūm al-Qurʾān Specialist (25%)

Phase 3: Pilot + Iteration (Months 7–12)

Objective: Deploy a closed pilot with a small scholarly user group; iterate based on feedback; complete the benchmark dataset; begin NLP model fine-tuning.

Key Activities:

1. Expand scoring system to first 30 surahs
2. Complete benchmark dataset: 100 verses, 5 annotators, IAA ≥ 0.70
3. Fine-tune CAMeLBERT-CA on tafsīr corpus for semantic similarity and cross-reference detection
4. Ingest asbāb al-nuzūl data (al-Wāḥidī, al-Suyūṭī) and hadith-verse linkages (ṣaḥīḥ grades only)
5. Deploy closed pilot to 10–15 beta users (mix of scholars, graduate students, interested laypeople)
6. Implement NBFI algorithm and test with real user interpretations
7. Red-team the system: attempt to produce false Green-band scores for known weak interpretations
8. Iterate on plain-language flag templates based on user feedback
9. Test two-pass scoring on the 100-verse benchmark
10. Measure divergence frequency across the benchmark
11. Calibrate the ≥25pt threshold using pilot data
12. (v2.1) Build and deploy Citation Verification Gate: implement source ID index, passage-match fuzzy search, and NLI claim-source alignment classifier
13. (v2.1) Build hallucination detection benchmark: 200+ adversarial false claims across all RDG categories; validate with scholars
14. (v2.1) Evaluate CVG against hallucination benchmark: target recall ≥ 95%, precision ≥ 90%

Deliverables:

- Scoring system covering 30 surahs
- Fine-tuned NLP models (evaluation report against benchmark)
- Closed pilot report (user feedback synthesis)
- Red-team report with documented failure modes
- Benchmark dataset v1.0 (locked)
- (v2.1) Citation Verification Gate deployed and integrated into NLP pipeline
- (v2.1) Hallucination detection benchmark v1.0 (locked, 200+ items, scholar-verified)
- (v2.1) CVG evaluation report (precision/recall against hallucination benchmark)

Milestone Gate: System scores ≥ 80% agreement with human benchmark; pilot users rate system outputs as "informative or highly informative" ≥ 70%; red-team report reviewed and critical failures mitigated; (v2.1) CVG recall ≥ 95% on hallucination benchmark

Team Required: Full team active; Arabic NLP Scientist (100%), Product Designer joining (50%), Hadith Specialist joining (50%)

Phase 4: v1 Production (Months 13–18)

Objective: Complete coverage of all 114 surahs; public launch of v1; establish ongoing scholarly review process.

Key Activities:

1. Complete ingestion of all Tier 5 tafsīr works for all 114 surahs
2. Ingest Tier 6 lexicographic data (al-Rāghib al-Iṣfahānī, Lane integration for all Qur'anic roots)
3. Complete qirāʾāt database for all 10 canonical recitations
4. Deploy React + D3 frontend (knowledge graph visualization; interpretation submission workflow)
5. Launch v1 with public access under clear governance documentation
6. Establish scholar review board for ongoing quality maintenance
7. Publish methodology paper in academic venue (Journal of Qur'anic Studies or equivalent)

Deliverables:

- v1 production system: all 114 surahs, full scoring API, public frontend
- Qirāʾāt database v1.0
- Governance documentation and public-facing methodology explainer
- Academic publication submitted

Milestone Gate: All 114 surahs covered with ≥ 3 Tier 5 sources; public launch with governance documentation; ≥ 3 Islamic scholars willing to be named advisors

Team Required: Full team; Community Scholar Liaison critical in this phase (100%)

Phase 5: Scale + Governance (Months 19–24)

Objective: Expand source coverage, improve NLP quality, establish sustainable governance, and publish findings.

Key Activities:

1. Ingest Tier 4 sources systematically (tābiʿūn literature from OpenITI)
2. Expand Tier 7 source coverage (modern academic scholarship, Fazlur Rahman, Seyyed Hossein Nasr, Angelika Neuwirth)
3. Fine-tune dedicated models for rhetorical device detection and asbāb al-nuzūl relevance scoring

4. Establish community contribution workflow: scholars can submit Tafsir_Claim nodes with full provenance; editorial review before ingestion
5. Add Shīʿī Imāmī tafsīr coverage (Tafsīr al-Mīzān — ʿAllāma Ṭabāṭabāʾī) as explicitly-labeled parallel track
6. Publish benchmark dataset and evaluation methodology publicly
7. Establish annual scholar review cycle; revise methodology document based on accumulated experience

Deliverables:

- Expanded corpus (Tier 3–4 and Tier 7 sources indexed)
- Shīʿī tafsīr parallel track
- Community contribution platform
- Published benchmark dataset
- Annual governance report

Milestone Gate: System covers ≥ 5 sources per tier for all tiers 3–7; community board established with at least 6 active scholars; published peer-reviewed methodology paper

Team Required: Smaller core team + expanded scholar advisory board; operations and community management roles added

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Section 10: Worked Example — Full Interpretation Evaluation

10.1 The Test Case

Verse: Qur'an 2:30

إِنِّي جَاعِلٌ فِي الْأَرْضِ خَلِيفَةً :

Transliteration: Innī jāʿilun fī l-arḍi khalīfatan

Standard translation (Pickthall): "Lo! I am about to place a viceroy in the earth."

User-submitted interpretation:

"Khalīfa here means political caliph — a ruler with divine authority to govern all Muslims."

10.2 Pre-Scoring: Verse Context Retrieval

Before scoring begins, the system retrieves:

From the Knowledge Graph:

- Surah 2 (al-Baqara): Madanī; verse 30 is in the opening section discussing creation and the human role
- Surrounding context: Verse 30 opens with God's address to the angels; angels ask "Will You place therein one who will cause corruption and shed blood?"; God responds "I know what you do not know" → humanity's knowledge of names (v.31) → angels' prostration (v.34)
- Qirāʾāt: No meaning-altering variant for khalīfatan in this verse across the canonical recitations (Ḥafṣ, Warsh, Qālūn all read the same form)
- Asbāb al-nuzūl: No specific occasion of revelation for this verse; it is part of the foundational Madanī narrative prologue

خلف:

- Root: - - (kh-l-f)
- Semantic fields documented in classical lexica:
  - Succession / coming after: "that which comes after another" (Lisān al-ʿArab: "al-khalaf: what comes after the salaf; al-khalīfa: the one who succeeds another in his place")

- Stewardship / deputyship: acting in place of, managing on behalf of (al-Mufradāt of al-Rāghib al-Iṣfahānī: "al-khalīfa: one who takes the place of another, either because the one replaced has departed, or because the one who places him honors him")
- Political ruler (historical usage): the title adopted by Abū Bakr al-Ṣiddīq as khalīfat rasūl Allāh (successor to the messenger of God) — this is a historical application that postdates the verse's revelation

- Lane's Lexicon on khalīfa: "A successor; a vicegerent; a deputy... khalīfat Allāh: God's vicegerent, His deputy on earth" — Lane notes the political usage explicitly but as a derived application of the succession/deputyship sense

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10.3 Scoring — Dimension by Dimension

Dimension 1 — Lexical Fidelity

Analysis:

خلف

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1. The form in the verse is khalīfatan — indefinite singular noun (ism, accusative, tanwīn). This is not the form used for a specific political title (al-khalīfa, definite) but for a category or function.

2. The primary sense of the root is succession/coming after or deputyship — not an inherent claim to divine authority to rule. Al-Rāghib al-Iṣfahānī in al-Mufradāt fī Gharīb al-Qurʾān — the specialized Qur'anic vocabulary lexicon — explicitly defines khalīfa here as "one who acts as deputy" without any political administration nuance being primary.

3. The phrase "divine authority to govern all Muslims" is not derivable from the text. The verse does not mention Muslims; it predates the Islamic community. The text says "in the earth" (fī l-arḍ) — a universal, not a specifically Islamic, designation.

4. The word khalīfatan is used in 2 explicit Qur'anic instances in this sense: Q2:30 (Ādam) and Q38:26 (Dāwūd, addressed directly by God). Neither context explicitly frames the role as political governance over a religious community.

D1 Score: 35/100

Flag issued: The lexical range of khalīfa includes succession and deputyship — not specifically political caliphate. The political caliphate title is a historically derived usage that postdates this verse. The phrase "govern all Muslims" has no lexical basis in this text.

Dimension 2 — Contextual Coherence

Analysis:

1. Surrounding verses: The immediate context (vv. 30–33) describes God placing Ādam in the earth, the angels' concern about human corruption, and God demonstrating Ādam's knowledge of names. The context is cosmological and anthropological — about humanity's nature, role, and accountability — not about political governance.

2. The angels' objection: "Will You place therein one who will cause corruption and shed blood?" is incoherent if khalīfa means political caliph. The angels are not objecting to political administration; they are raising a question about the purpose of placing humans on earth at all.

3. The answer: God's response references knowledge ("I know what you do not know") and demonstrates Ādam's capacity to name all things. This is an argument about human cognitive and stewardship capacity, not about political legitimacy.

4. Makkī/Madanī context: This is a Madanī surah, but this passage is part of a narrative prologue that deals with universal themes of creation and human nature, not specifically with Muslim political organization.

5. The phrase "all Muslims" directly fails contextual coherence: the verse is set at the beginning of human existence, before any Islamic community existed.

D2 Score: 15/100

Flag issued: The surrounding context (vv. 30–38) is about human nature, stewardship, and accountability on earth — not political governance. The angels' question and God's response make no sense if the referent is a political caliphate. The phrase "govern all Muslims" is temporally and contextually impossible given the verse's setting.

Dimension 3 — Transmitted Support

Analysis:

The system retrieves all Tier 1–4 evidence for Q2:30, khalīfa interpretation:

Tier 1 (Qur'an-by-Qur'an):

- Q38:26 — God says to Dāwūd: "O Dāwūd, We have made you a khalīfa in the earth, so judge between people in truth and do not follow desire." This cross-reference uses the same term for an individual with a mandate of just governance and accountability — which constrains the sense toward vicegerency/stewardship with accountability, not toward an unrestricted political authority over all Muslims.

Tier 2 (Prophetic):

- No specific ṣaḥīḥ hadith directly explains the word khalīfa in 2:30 as "political caliphate." The absence is significant; if the political sense were primary, Prophetic tafsīr would likely have

asserted it.

Tier 3 (Companions):

- Ibn ʿAbbās (as reported in al-Ṭabarī): "khalīfatan" means "one who succeeds others (mankind before Ādam) in the earth" — focusing on the succession sense. Ibn ʿAbbās does not use political governance as the primary sense.
- ʿAlī ibn Abī Ṭālib: no specific transmitted interpretation of this word in this verse in the major collections that frames it as political caliphate.

Tier 4 (Successors):

- Mujāhid ibn Jabr: interpreted khalīfa as "a deputy (nāʾib) of God in applying His commands" — stewardship/accountability language, not political rulership of a religious community.
- Al-Ḍaḥḥāk: "they are called khulafāʾ because they manage (yaʿmurūn) the earth" — again, a stewardship reading.

Assessment: The "political caliphate with divine authority over all Muslims" interpretation has no Tier 1–4 support. The transmitted tradition consistently reads khalīfa here as stewardship, succession, or deputyship — not as a post-prophetic political title.

D3 Score: 10/100

Flag (Critical): No Tier 1, 2, 3, or 4 source supports reading khalīfa in 2:30 as "political caliph with divine authority over all Muslims." The transmitted record uniformly reads it as stewardship, succession, or deputyship. This triggers the hard-cap rule for D3.

Dimension 4 — Scholarly Precedent

Analysis:

The system searches all indexed Tier 5 sources for Q2:30 khalīfa interpretation:

- Al-Ṭabarī (Jāmiʿ al-Bayān, Vol. 1 on 2:30): Presents multiple readings — primarily successor/steward; notes the angels' concern implies a general human role on earth. Al-Ṭabarī does not read this verse as establishing a political institution.
- Ibn Kathīr (Tafsīr al-Qurʾān al-ʿAẓīm, Vol. 1 on 2:30): Explains khalīfa as "those who succeed one another in the earth, generation by generation and century by century" — explicitly generational succession, not a single political title. Ibn Kathīr cites Ibn ʿAbbās and Mujāhid.
- Al-Qurṭubī (al-Jāmiʿ li-Aḥkām al-Qurʾān, Vol. 1 on 2:30): Explicitly addresses competing readings. He discusses whether khalīfa means (a) one who succeeds those before him, (b) God's deputy (khalīfat Allāh), or (c) the human successor to other creatures on earth. Al-Qurṭubī does not advance the "political caliph ruling all Muslims" reading; his analysis focuses on accountability and stewardship. He does briefly note the political use of the term but

distinguishes it from this verse's context.

• Al-Zamakhsharī (al-Kashshāf, on 2:30): From a Muʿtazilī perspective, emphasizes human rational agency and accountability; khalīfa is about delegated stewardship.

• Ibn ʿĀshūr (al-Taḥrīr wa'l-Tanwīr): Modern classical synthesis; khalīfa is humanity's stewardship and accountability role on earth; Ibn ʿĀshūr explicitly resists readings that narrowly equate the term with any specific political institution.

The specific user interpretation ("political caliph... divine authority to govern all Muslims") does not appear in any major classical tafsīr. This specific formulation appears to be a modern retroactive reading — projecting the later political caliphate institution backward onto a pre-Islamic text.

D4 Score: 5/100

Flag (Critical): No major classical tafsīr supports the specific interpretation submitted. Al-Qurṭubī mentions the political usage but does not apply it to this verse. This is identified as a Category 4 (speculative/modern) claim that has no classical scholarly precedent.

Dimension 5 — Internal Consistency

Analysis:

If khalīfa in 2:30 means "political caliph with divine authority to govern all Muslims," this creates at least two internal consistency problems:

1. Q38:26 uses the same term for Dāwūd, commanding him to judge with justice — not claiming authority over "all Muslims" (who did not exist then either). If khalīfa means "political caliph of all Muslims," then Q38:26 becomes incoherent (Dāwūd as caliph of all Muslims?). This interpretation forces an incoherent reading on Q38:26.

2. Q2:30's narrative logic: The verse introduces Ādam specifically, and the angels express concern about human nature. If Ādam is a political caliph of all Muslims, why are the angels concerned about corruption and bloodshed? A political title does not logically connect to the cosmological concern being raised.

3. Semantic scope: "All Muslims" is a temporally limited category; the verse's wording is cosmologically unlimited ("in the earth" — fī l-arḍ). Narrowing to "all Muslims" requires ignoring the universalist language.

D5 Score: 30/100

Flag: This interpretation creates an internal inconsistency with Q38:26 (where the same term applies to Dāwūd with no plausible "caliph of all Muslims" sense) and with the narrative logic of the surrounding passage.

10.4 Composite Scoring (v2.0 — Two-Pass)

Pass 1: Textual Logic Score (TLS)

TLS uses D1, D2a, D2b, and D5 only (no external tradition).

For this example:

- D2a (Thematic Context): 15/100 — The surrounding narrative is cosmological/anthropological, not political. The angels' question and God's response center on human nature, not governance.
- D2b (Temporal Context): 20/100 — No temporal error per se (no later verse supersedes this), but the interpretation projects a post-prophetic political institution backward into a pre-Islamic cosmological passage — a temporal category error.

Dimension	Score	TLS Weight	TLS Contribution
D1 — Lexical Fidelity	35	0.35	12.25
D2a — Thematic Context	15	0.15	2.25
D2b — Temporal Context	20	0.15	3.00
D5 — Internal Consistency	30	0.35	10.50
TLS (Pass 1)			28.0

Pass 2: Integrated Tradition Score (ITS)

ITS uses all six dimensions including D3 and D4.

Dimension	Score	ITS Weight	ITS Contribution
D1 — Lexical Fidelity	35	0.20	7.00
D2a — Thematic Context	15	0.10	1.50
D2b — Temporal Context	20	0.10	2.00
D3 — Transmitted Support	10	0.30	3.00
D4 — Scholarly Precedent	5	0.20	1.00
D5 — Internal Consistency	30	0.10	3.00

Dimension	Score	ITS Weight	ITS Contribution
ITS (Pass 2)			17.5

Divergence Check: |TLS − ITS| = |28.0 − 17.5| = 10.5 → No Structural Divergence Warning triggered.

Confidence Score: 82 / 100 (Verse 2:30 is well-covered: multiple Tier 1–5 sources, high IAA among annotators)

Result:

- TLS: 28 / 100 — Red band: the text itself does not support this reading
- ITS: 17.5 / 100 — Red band: tradition firmly contradicts this reading
- Verdict: Red — Unsupported / Contradicted by both text and tradition

Both scores are low and close together: this is an interpretation that fails on text-internal grounds and tradition grounds simultaneously. No divergence warning is needed — the system and the tradition agree on the verdict.

10.5 Plain-Language Misalignment Explanation (User-Facing)

Your interpretation: "Khalīfa here means political caliph — a ruler with divine authority to govern all Muslims."

Your score: 17/100 — Unsupported / Contradicted

Here is why, dimension by dimension:

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1. The Arabic word (Lexical Fidelity — 35/100)

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2. Does it fit the context? (Contextual Coherence — 15/100)

The scene in this verse is God speaking to the angels before human beings existed. The angels respond by asking: "Will you place in it one who causes corruption and sheds blood?" — a question about human nature and purpose on earth, not about who should run a government. God's response is about wisdom and knowledge, not about political authority. Also: the phrase "all Muslims" makes no sense in this context — there were no Muslims when Adam was created. The verse uses the phrase "in the earth" (fī l-arḍ) — which covers all of humanity, not a specific religious community.

3. What the earliest sources say (Transmitted Support — 10/100)

We checked the earliest Islamic scholarship on this verse, from the Prophet's Companions and their students:

- Ibn ʿAbbās (the Prophet's cousin, the most cited Companion interpreter): explained khalīfa as the human role of succession on earth — humans succeeding those who came before.
- Mujāhid ibn Jabr (a leading early scholar): interpreted it as acting as God's deputy in applying His commands — a stewardship reading.
- No Companion, no early scholar, and no authenticated hadith supports the specific reading of "political caliph with divine authority over all Muslims."

This is the most important failure in your interpretation. The Islamic scholarly tradition has a strong principle: an interpretation that has no support from early sources requires extraordinary justification.

4. What classical scholars say (Scholarly Precedent — 5/100)

We checked the five most widely used classical commentaries:

- Al-Ṭabarī (d. 923 CE): khalīfa = human beings succeeding each other on earth, generation by generation.
- Ibn Kathīr (d. 1373 CE): khalīfa = generations succeeding one another; not a specific political title.
- Al-Qurṭubī (d. 1273 CE): acknowledges multiple meanings but does not apply "political caliph" to this verse.
- Al-Zamakhsharī (d. 1144 CE): focuses on stewardship and accountability.
- Ibn ʿĀshūr (d. 1973 CE): explicitly warns against narrowing this verse to any specific political institution.

None of these scholars support your interpretation.

5. Does it fit the rest of the Qur'an? (Internal Consistency — 30/100)

The same word khalīfa appears in Qur'an 38:26, where God says to Dāwūd (David): "O Dāwūd, We have made you a khalīfa in the earth — judge between people with truth." If khalīfa means "caliph of all Muslims," then Dāwūd (who lived 1000 years before Islam) would be "the caliph of all Muslims" — which makes no sense. This shows that khalīfa in the Qur'an is not a political title for a leader of Muslims, but a stewardship role that can apply to prophets and to humanity as a whole.

10.6 Nearest Better-Framed Interpretation

Based on the knowledge graph retrieval and NBFI algorithm, the three closest well-supported alternatives are:

Alternative 1 (Score: 80/100 — Green)

"Khalīfa here refers to Ādam as the first human being, placed on earth as a steward and successor to whatever creatures preceded him, accountable to God for how he manages the earth."

- Supported by: Ibn ʿAbbās (Tier 3), Mujāhid (Tier 4), al-Ṭabarī and Ibn Kathīr (Tier 5)
- Why it scores higher: It honors the succession/stewardship sense of the root, fits the narrative context, and is directly supported by the earliest transmitted scholarship
- How it differs from yours: The role is about stewardship and accountability, not political governance; it applies to Ādam specifically as a human being, not to a post-prophetic political institution

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Alternative 2 (Score: 73/100 — Yellow)

"Khalīfa refers to humanity collectively as God's vicegerents on earth — trusted stewards of creation who are accountable to God for how they manage it."

- Supported by: multiple classical scholars including al-Qurṭubī; prominent in Ibn ʿĀshūr's synthesis (Tier 5); coherent with the Qur'anic principle of human accountability (amāna)
- Why it scores higher: Honors the cosmological scale of the verse ("in the earth"), consistent with the angels' concern about human nature, consistent with Q38:26
- How it differs from yours: "All Muslims" → "all of humanity"; "divine authority to govern" → "stewardship and accountability"; no political governance dimension

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Alternative 3 (Score: 68/100 — Yellow)

"Khalīfa means that human beings succeed one another on earth, each generation inheriting the earth from the previous — a role of succession and continuing stewardship."

- Supported by: Ibn ʿAbbās (Tier 3), al-Ṭabarī's primary reading (Tier 5)
- Why it scores higher: Most directly aligned with the primary lexical meaning of the root (succession); supported by the earliest transmitted scholarship
- How it differs from yours: The succession is generational, not a single political title; there is no claim to divine authority over a specific religious community

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System note: Your interpretation's claim that khalīfa specifically means political caliph and that this implies divine authority to govern all Muslims is a reading that developed in later Islamic political thought as a retroactive justification for the Caliphate institution. It is not attested in the Qur'anic, Prophetic, Companion, or classical tafsīr evidence for this verse. The system recommends Alternative 1 or 2 as better-evidenced framings of the verse's teaching.

10.7 A Divergence Example

To illustrate a case where Pass 1 and Pass 2 diverge, consider:

Verse: 5:6 (the ablution verse — washing vs. wiping feet)

Interpretation: "This verse requires washing the feet during wuḍūʾ."

Dimension	Score	Notes
D1 (Lexical Fidelity)	75	The naṣb reading (arjulakum, accusative) supports washing; the jarr reading (arjulikum) supports wiping. Both are mutawātir.
D2a (Thematic Context)	80	Consistent with the ablution instruction passage
D2b (Temporal Context)	85	Late Madanī; no later verse modifies this ruling
D3 (Transmitted Support)	90	Multiple ṣaḥīḥ hadith confirm the Prophet washed his feet; Ibn ʿAbbās, Ibn Masʿūd confirm
D4 (Scholarly Precedent)	95	Four Sunnī schools unanimously require washing

Dimension	Score	Notes
D5 (Internal Consistency)	70	No internal Qurʾānic contradiction, but the mutawātir jarr reading creates genuine textual ambiguity

TLS (Pass 1): (75 × 0.35) + (80 × 0.15) + (85 × 0.15) + (70 × 0.35) = 26.25 + 12 + 12.75 + 24.5 = 75.5

ITS (Pass 2): (75 × 0.20) + (80 × 0.10) + (85 × 0.10) + (90 × 0.30) + (95 × 0.20) + (70 × 0.10) = 15 + 8 + 8.5 + 27 + 19 + 7 = 84.5

Divergence: |75.5 – 84.5| = 9 points → No Structural Divergence Warning triggered.

Verdict: Green — both text and tradition support this reading.

Now consider the alternative interpretation: "This verse requires wiping the feet only."

TLS: ~73 (the jarr reading is equally mutawātir; textually the case is almost symmetric)

ITS: ~38 (contradicts Tier 2 Prophetic practice and Tier 3 Companion consensus; supported only by Shīʿī Jaʿfarī school)

Divergence: |73 – 38| = 35 → Structural Divergence Warning triggered.

User sees: "This reading is textually coherent (both qirāʾāt are authentic) but departs from the majority transmitted Prophetic practice. The Qur'anic text alone does not resolve this dispute — both readings are mutawātir. The tradition adjudicates in favor of washing."

This is the system working as designed: the text is genuinely ambiguous, the tradition provides resolution, and the user sees exactly where and how.

Appendix A: Key Classical Sources Referenced

Source	Author	Death Date	Language	Role in This Framework
Jāmiʿ al-Bayān ʿan Taʾwīl Āy al-Qurʾān	Muḥammad ibn Jarīr al-Ṭabarī	310 AH / 923 CE	Arabic	Primary Tier 5 source; synthesizes Tiers 1–4 with critical evaluation
Tafsīr al-Qurʾān al-ʿAẓīm	Ibn Kathīr (Ismāʿīl ibn Kathīr)	774 AH / 1373 CE	Arabic	Primary Tier 5 source; strong hadith methodology; limited Isrāʾīliyyāt
al-Jāmiʿ li-Aḥkām al-Qurʾān	Muḥammad ibn Aḥmad al-Qurṭubī	671 AH / 1273 CE	Arabic	Primary Tier 5 source; jurisprudential emphasis; Mālikī school
al-Kashshāf	Maḥmūd ibn ʿUmar al-Zamakhsharī	538 AH / 1144 CE	Arabic	Tier 5; outstanding rhetorical analysis; Muʿtazilī theological school (must tag)
Mafātīḥ al-Ghayb	Fakhr al-Dīn al-Rāzī	606 AH / 1210 CE	Arabic	Tier 5; encyclopedic rationalist method; Ashʿarī theological school
al-Taḥrīr wa'l-Tanwīr	Muḥammad al-Ṭāhir ibn ʿĀshūr	d. 1973 CE	Arabic	Tier 5 (late); major modern classical synthesis; Mālikī school
al-Mufradāt fī Gharīb al-Qurʾān	Al-Rāghib al-Iṣfahānī	d. ~502 AH	Arabic	Tier 6; specialized Qur'anic vocabulary lexicon; essential for Dimension 1
Lisān al-ʿArab	Ibn Manẓūr	711 AH / 1311 CE	Arabic	Tier 6; comprehensive classical Arabic dictionary

Source	Author	Death Date	Language	Role in This Framework
An Arabic-English Lexicon	Edward William Lane	d. 1876 CE	English	Tier 6 (Western scholarship); most comprehensive Arabic-English lexicon; draws from classical sources
al-Itqān fī ʿUlūm al-Qurʾān	Jalāl al-Dīn al-Suyūṭī	911 AH / 1505 CE	Arabic	ʿUlūm al-Qurʾān reference; asbāb al-nuzūl compilation
Asbāb al-Nuzūl	ʿAlī ibn Aḥmad al-Wāḥidī	468 AH / 1075 CE	Arabic	Primary source for asbāb al-nuzūl data

Appendix B: Glossary of Technical Terms

Term	Brief Definition
Tafsīr	Qur'anic exegesis/commentary
Tafsīr bi'l-maʾthūr	Tradition-based interpretation using transmitted reports
Tafsīr bi'l-raʾy	Reason-based interpretation using disciplined scholarly judgment
Isnād	Chain of transmission for a hadith
Asbāb al-nuzūl	Occasions/circumstances of revelation
Qirāʾāt	Variant recitations of the Qur'an
Naskh	Abrogation — later revelation superseding an earlier ruling
Muḥkam	Clear, unambiguous verse
Mutashābih	Ambiguous verse open to interpretation
Makkī	Revealed before the migration to Medina (~610–622 CE)
Madanī	Revealed after the migration to Medina (~622–632 CE)
Ṣaḥīḥ	Sound/authentic hadith grade
Ḥasan	Good/acceptable hadith grade
Ḍaʿīf	Weak hadith grade
Mawḍūʿ	Fabricated hadith — not a legitimate source
Ṣaḥāba	Companions of the Prophet
Tābiʿūn	Successors — the generation that learned from the Companions
Ikhtilāf	Documented scholarly disagreement
Ijmāʿ	Scholarly consensus
Marjūḥ	Outweighed/minority scholarly position

Term	Brief Definition
Isrā’īliyyāt	Narratives from Jewish or Christian tradition that entered some tafsīr
Balaghah	Arabic rhetoric and eloquence
ʿUlūm al-Qur’ān	Sciences of the Qur'an (the collective disciplines of Qur'anic study)
Madhab	Legal school (Ḥanafī, Mālikī, Shāfiʿī, Ḥanbalī)
ʿAqīdah	Islamic theology / creedal belief
Ijtihād	Independent legal or interpretive reasoning by a qualified scholar
Mujtahid	One qualified to perform ijtihād
Mufassir	Qur'anic exegete/commentator
Khalīfa	Successor; deputy; steward; vicegerent (context-dependent)
RDG (Retrieve-Don't-Generate)	Architectural principle (v2.1): any factual claim about external sources (lexicons, hadith, tafsīr, scholarly positions) must be retrieved from indexed corpora, never generated from LLM parametric memory. See Section 5c-1.
AI-as-Analyst	RDG category: AI operations that compute directly on visible data (morphological analysis, embeddings, pattern detection, scoring). Permitted without retrieval constraint.
AI-as-Scholar-Proxy	RDG category: AI operations that assert what a human source said. Must always retrieve from indexed sources; never generate from memory.
Citation Verification Gate (CVG)	Three-step automated verification pipeline (v2.1) that intercepts all Scholar-Proxy claims before graph ingestion. Steps: source ID lookup, passage match (≥ 0.85 fuzzy threshold), claim-source NLI alignment (≥ 0.70 confidence). See Section 5e.
NLI (Natural Language Inference)	Classification task determining whether a hypothesis is entailed by, contradicts, or is neutral to a premise. Used in CVG Step 3 to verify that derived conclusions follow from cited source passages.

End of Document

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